CHY1008
Basic Chemistry and Environmental Studies

Module 4

Part 1

Engineering Materials

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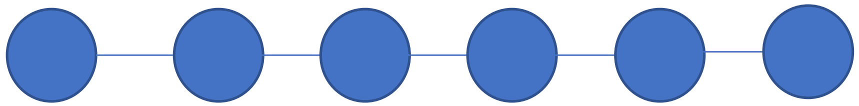
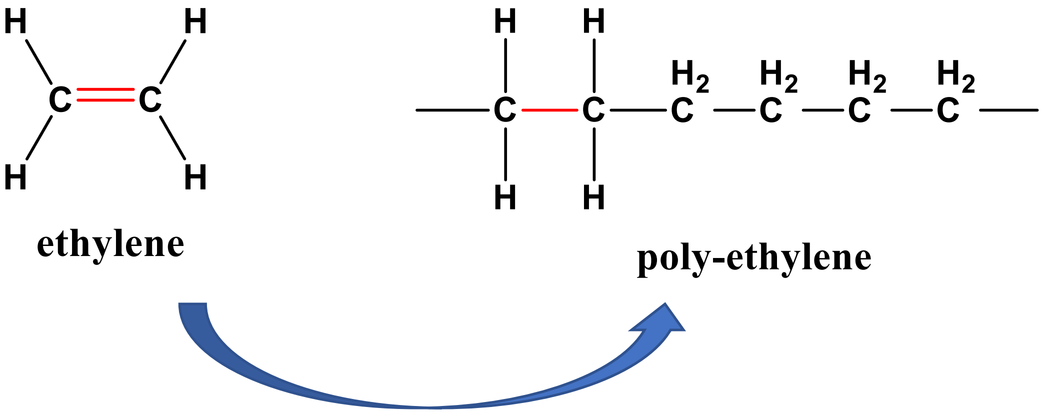
Polymers

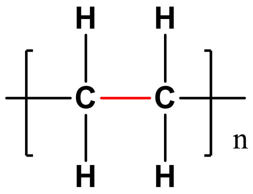
Monomer & Polymer

- The monomer is the *repeat unit* of the polymer





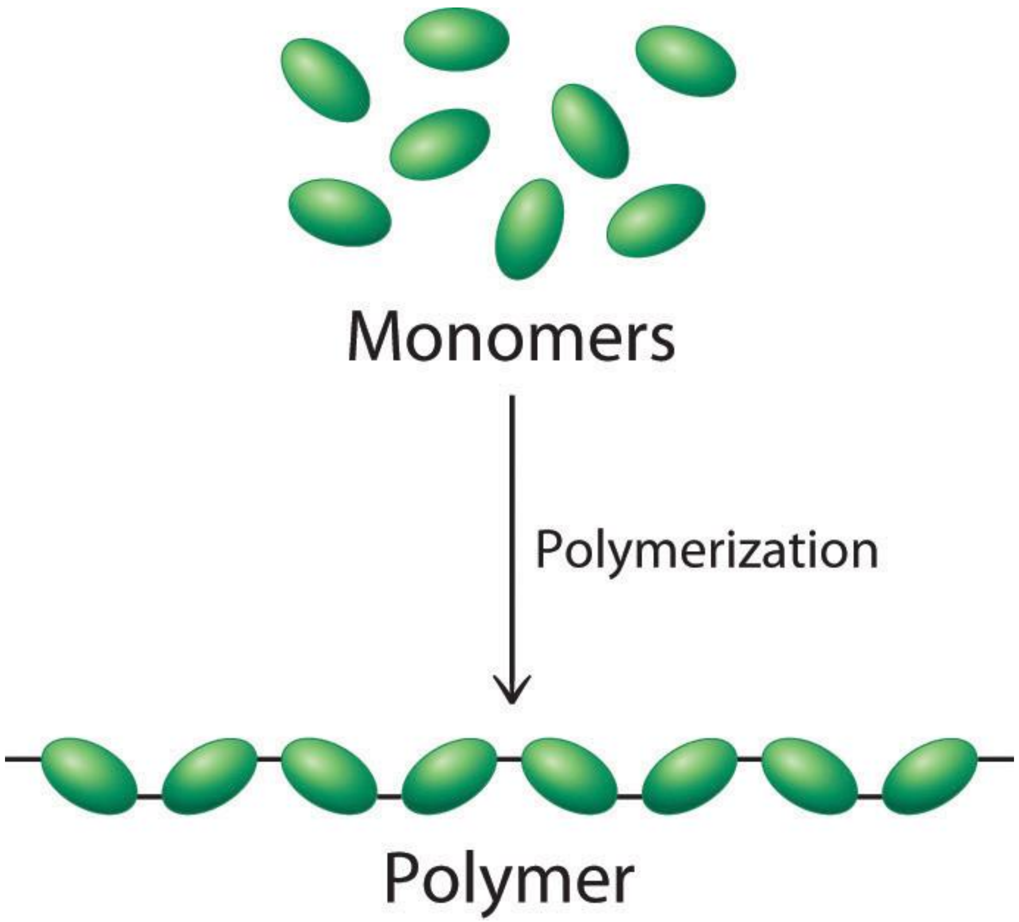
Polymerization Reaction



Polymer

- The word polymer is derived from the classical Greek words **poly** meaning "**many**" and **meres** meaning "**parts**."

- Simply stated, **a polymer is a long-chain molecule that is composed of a large number of repeating units of identical structure.**



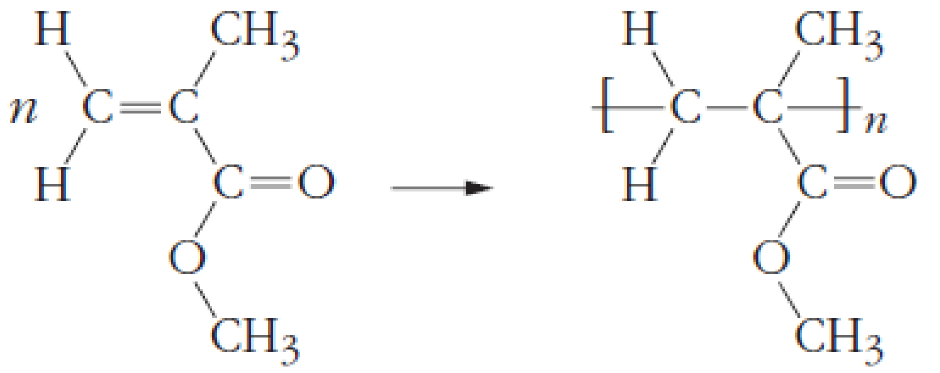
Polymer

- The **monomer** is the *repeat unit* of the polymer, and a typical polymer may have from hundreds to *hundreds of thousands of repeat units*

- *Synthetic* polymers are created by chemical reactions in the laboratory

- *Natural* polymers (or *biopolymers*) are created by chemical reactions within organisms.

Polymerization Reaction



Methyl methacrylate monomer

Poly(methyl methacrylate)

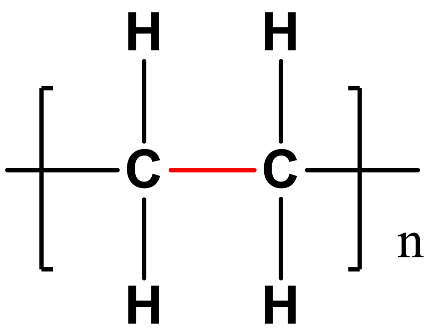
- ✓ *n-monomers are added* together to form a polymer chain with ***n-repeat units.***
- ✓ ***molecular weight of a polymer is an average quantity***
- ✓ ***High Molecular weight***
- ✓ Mechanical properties, *toughness, rigidity, plasticity*, and their *thermal softening* properties

Degree of polymerization (DP)

- Degree of polymerization is defined as the ratio between the molar mass of the polymer and molar mass of the repeat unit.

- It gives the value of '**n**' $nA \rightarrow (A)_n$

$$\text{Degree of polymerization} = \frac{\text{molar mass of polymer}}{\text{molar mass of monomer}}$$



Molecular Mass of Polymer

Even though any *given* chain within a sample of a polymer has a fixed molar mass, the **degree of polymerization often varies considerably from chain to chain**.

For this reason, polymer chemists use various definitions of ***average molar mass***, and a common one is the ***number-average molar mass***:

Table 12.8 Molar Masses of Some Common Polymers

Name	$\mathcal{M}_{polymer}$ (g/mol)	n	Uses
Acrylates	2×10^5	2×10^3	Rugs, carpets
Polyamide (nylons)	1.5×10^4	1.2×10^2	Tires, fishing line
Polycarbonate	1×10^5	4×10^2	Compact discs
Polyethylene	3×10^5	1×10^4	Grocery bags
Polyethylene (ultrahigh molecular weight)	5×10^6	2×10^5	Hip joints
Poly(ethylene terephthalate)	2×10^4	1×10^2	Soda bottles
Polystyrene	3×10^5	3×10^3	Packing, coffee cups
Poly(vinyl chloride)	1×10^5	1.5×10^3	Plumbing

Molecular Mass of Polymer

1. Number average Molecular Mass (M_n)

$$\mathcal{M}_n = \frac{\text{total mass of all chains}}{\text{number of moles of chains}}$$

$$\bar{M}_n = \frac{\sum_{i=1}^{N} N_i M_i}{\sum_{i=1}^{N} N_i}$$

$$M_n = \frac{M_1 n_1 + M_2 n_2 + M_3 n_3 + \cdots}{n_1 + n_2 + n_3 + \cdots}$$

2. Weight Average Molecular Mass (M_w)

$$\bar{M}_w = \frac{\sum_{i=1}^{N} N_i M_i^2}{\sum_{i=1}^{N} N_i M_i} = \frac{\sum_{i=1}^{N} W_i M_i}{\sum_{i=1}^{N} W_i}$$

$$M_w = \frac{M_1^2 \quad n_1 + M_2^2 \quad n_2 + M_3^2 \quad n_3 \ldots}{n_1 M_1 + n_2 M_2 + n_3 M_3 + \cdots}$$

QN 1 Calculation of Number Average Molecular Mass

S.No	Number of Molecules	Molecular Weight gm/mol
1	2	5
2	3	10
3	4	20
4	10	40

QN 2 Calculate the weight-average molar mass

S.No	Number of Molecules	Molecular Weight gm/mol
1	2	5
2	3	10
3	4	20
4	10	40

M_w is always greater than M_n

$M_w > M_n$

QN

Calculate M_n and M_w

No. of moles of i^{th} polymeric chain (N_i)	Molar mass of i^{th} polymeric chain (M_i)
250	12500
325	14520
175	16500
200	17250

$$\bar{M}_n = \frac{\sum_{i=1}^{N} N_i M_i}{\sum_{i=1}^{N} N_i}$$ **=14927.89**

$$\bar{M}_w = \frac{\sum_{i=1}^{N} N_i M_i^2}{\sum_{i=1}^{N} N_i M_i}$$ **=15142.17**

PDI=1.014

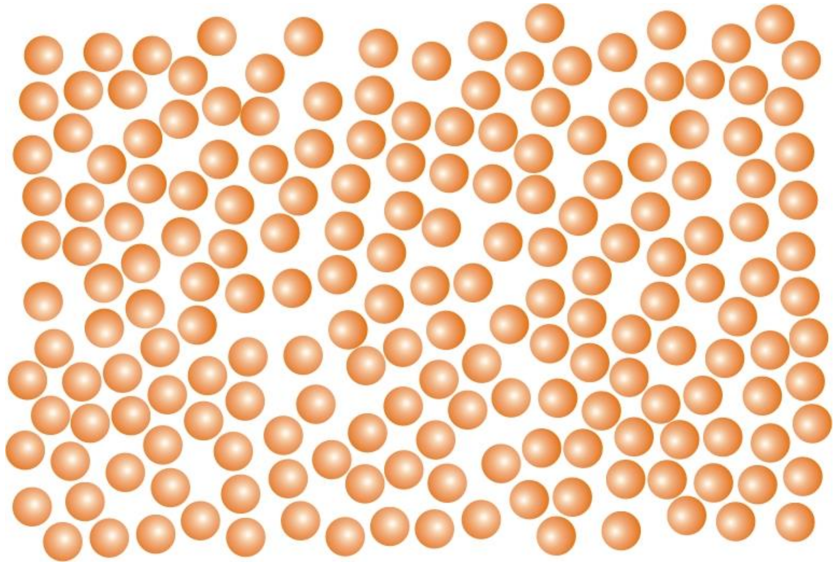
Polydispersive Index (PDI)

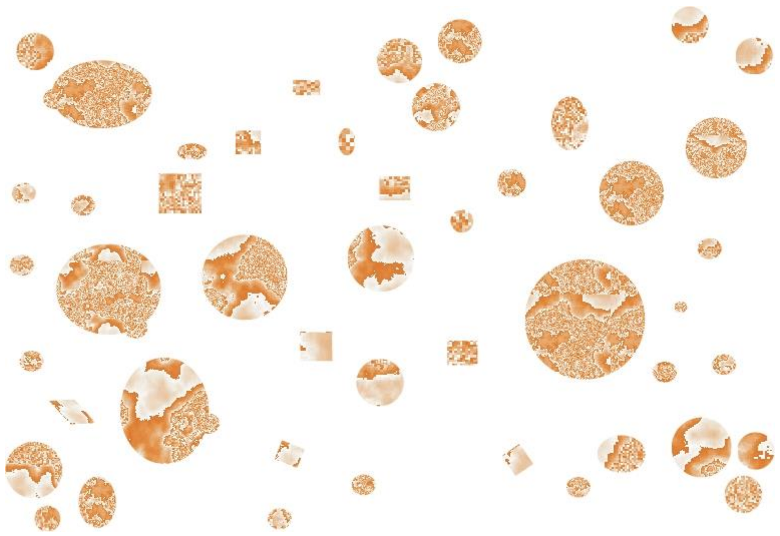
- Ratio of M_w and M_n is known as Polydispersity Index (PDI)

$$\mathrm{PDI} = \frac{M_w}{M_n}$$

- PDI = 1 (Monodisperse) Ex: *DNA, RNA, Proteins* $M_w = M_n$

- PDI > 1 (Polydisperse) Ex: Synthetic Polymers $M_w > M_n$





QN Calculate the PDI value

No. of moles of i^{th} polymeric chain (N_i)	Molar mass of i^{th} polymeric chain (M_i)
250	12500
325	14520
175	16500
200	17250

$$\bar{M}_n = \frac{\sum_{i=1}^{N} N_i M_i}{\sum_{i=1}^{N} N_i}$$ **=14927.89**

$$\bar{M}_w = \frac{\sum_{i=1}^{N} N_i M_i^2}{\sum_{i=1}^{N} N_i M_i}$$ **=15142.17**

QN A polymer has polydispersity index of 1.98. Find out the weight average molecular weight (M_w) of the polymer from the following data:

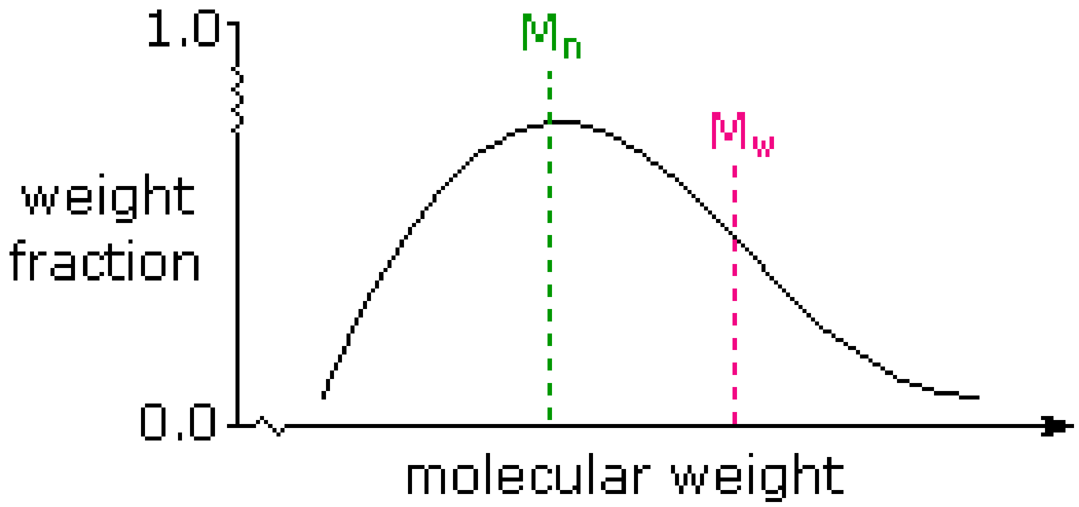
No. of different types of molecules (N_i)	Molar mass of each type of molecules (M_i)
150	82500
230	95000
355	35000
490	40200

$$\bar{M}_n = \frac{\sum_{i=1}^{N} N_i M_i}{\sum_{i=1}^{N} N_i} = \mathbf{54161.63}$$

$$PDI = \frac{M_w}{M_n}$$

$$\mathbf{M_w = 107240.03}$$

M_w is always greater than M_n



M_n and M_w in a typical sample
of polydispersed macromolecules

$$M_w > M_n$$

Problem 1

A polydisperse sample of polystyrene is prepared by mixing three *monodisperse* samples in the following proportions:

1 g	10,000 molecular weight
2 g	50,000 molecular weight
2 g	100,000 molecular weight

Using this information, calculate the number-average molecular weight, weight-average molecular weight, and PDI of the mixture.

Problem 1

A polydisperse sample of polystyrene is prepared by mixing three *monodisperse* samples in the following proportions:

1 g	10,000 molecular weight
2 g	50,000 molecular weight
2 g	100,000 molecular weight

Using this information, calculate the number-average molecular weight, weight-average molecular weight, and PDI of the mixture.

$$\bar{M}_\mathrm{n} = \frac{\sum_{i=1}^{3} N_i M_i}{\sum_{i=1}^{3} N_i} = \frac{\sum_{i=1}^{3} W_i}{\sum_{i=1}^{3} \left(W_i / M_i\right)} = \frac{1+2+2}{\frac{1}{10,000}+\frac{2}{50,000}+\frac{2}{100,000}} = 31,250$$

$$\bar{M}_\mathrm{w} = \frac{\sum_{i=1}^{3} N_i M_i^2}{\sum_{i=1}^{3} N_i M_i} = \frac{\sum_{i=1}^{3} W_i M_i}{\sum_{i=1}^{3} W_i} = \frac{10,000+2(50,000)+2(100,000)}{5} = 62,000$$

$$\mathrm{PDI} = \frac{\bar{M}_\mathrm{w}}{\bar{M}_\mathrm{n}} = \frac{62,000}{31,250} = 1.98$$

Problem 2

A polymer has the following composition: 100 molecules of molecular mass 1000 g/mol, 200 molecules of molecular mass 2000 g/mol and 500 molecules of molecular mass 5000 g/mol. Calculate the number and weight average molecular weight and the polydispersity index.

Problem 2

Given that $M_1 = 1000$ g/mol, $N_1 = 100$; $M_2 = 2000$ g/mol, $N_2 = 200$; $M_3 = 5000$ g/mol, $N_3 = 500$.
The number average molecular weight in given by:

$$\overline{M_n} = \frac{\sum N_i M_i}{\sum N_i} = \frac{100\times 1000 + 200\times 2000 + 500 + 5000}{100 + 200 + 500} = \frac{1\times 10^5 + 4\times 10^5 + 25\times 10^5}{800}$$

$$= 3.75\times 10^3 \text{g/mol}$$

The weight average molecular weight is given by:

$$\overline{M_w} = \frac{\sum N_i M_i^2}{\sum N_i M_i} = \frac{100\times (1000)^2 + 200\times (2000)^2 + 500\times (5000)^2}{30\times 10^5}$$

$$= \frac{1\times 10^8 + 8\times 10^8 + 125\times 10^8}{30\times 10^5} = \frac{134\times 10^8}{30\times 10^5} = 4.46\times 10^3 \text{ g/mol}$$

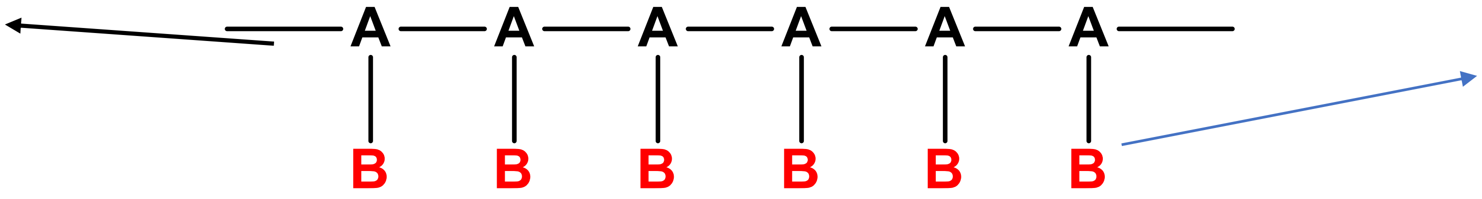
The polydispersity index (PDI) is

$$\frac{\overline{M_w}}{\overline{M_n}} = \frac{4.46\times 10^3}{3.75\times 10^3} = 1.19$$

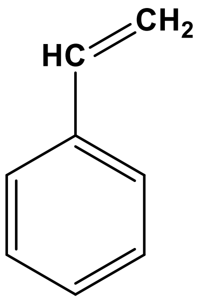
Structure of Polymer

Each polymer chain will have a **backbone** and may have **pedant** groups (side chain or branching atoms).

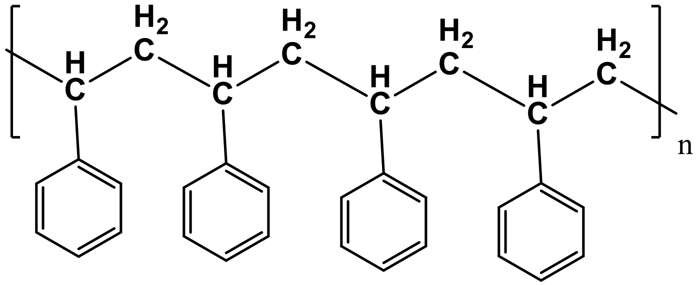
A - Backbone



B - **pedant** groups



Styrene



Poly-Styrene

Phenyl group is the pendant group

Classification of Polymers

Classification of Polymers

Chemical Structure

- Homopolymer — Consists of identical monomers
- Copolymer — Mixture of more than one type of monomer

Polymeric structure

Tacticity

Thermal Behaviour

Molecular Force

Methods of Synthesis

Origin

Copolymer

Polymers with two different repeating units in their chains are called **copolymers**

—A—A—A—A—A—A— **Polymer**

—A—B—A—B—A—B—A—B—A—A— **Regular Copolymer**

—A—A—B—B—A—B—A—A—A—B— **Random Copolymer**

—A—A—A—A—B—B—A—A—B—B—B—A—A—A—A— **Block Copolymer**

```
               B—B—B—B
               |
—A—A—A—A—A—A—A—A—A—A—A—A—
                     |
                     B—B—B—B
```

Graft Polymer

[...—A—B—B—A—A—A—B—A—B—B—...]

Random copolymer

[...—A—B—A—B—A—B—A—B—A—B—...]

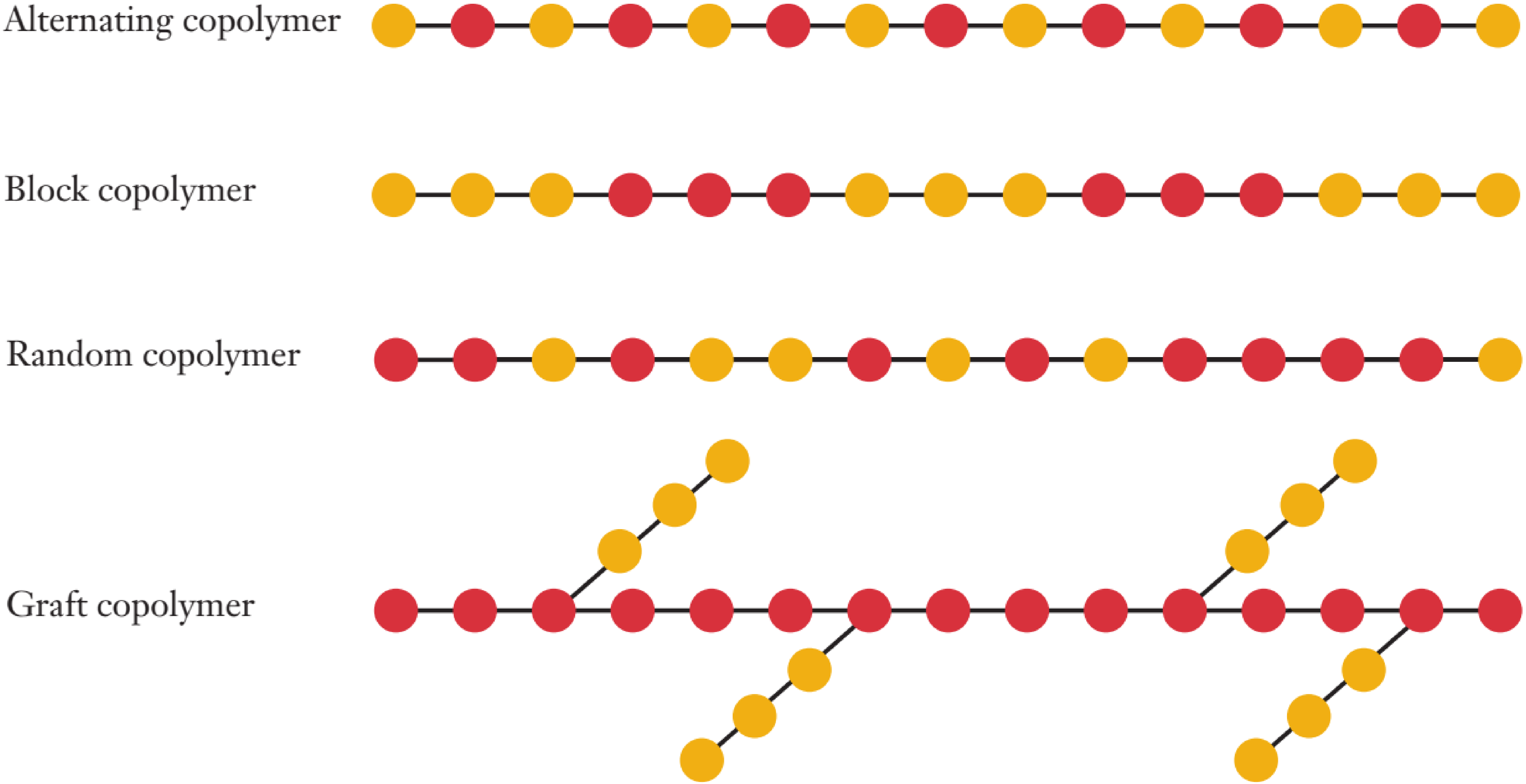
Regular copolymer

[...—A—A—A—A—B—B—B—B—A—A—B—B—B—B—B—B—A—A—A...]

Block copolymer

```
[ ...—A—A—A—A—A—A—A—A—A—A—... ]
            |
            B—B—B—B—B—...
```

Graft copolymer


Alternating copolymer
Block copolymer
Random copolymer
Graft copolymer

Block Polymers

- **AB-block copolymers:** Copolymers that contain a long block of one monomer (A) followed by a block of the other monomer (B).

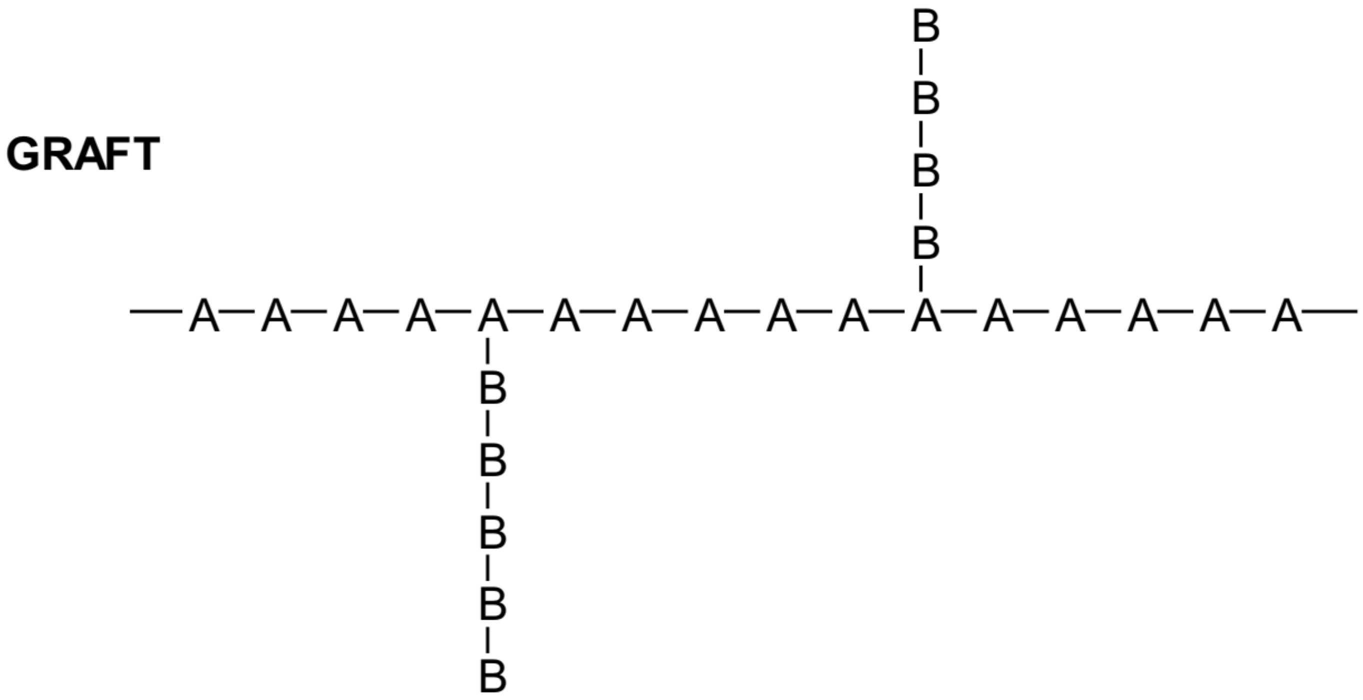
-A-A-A-A-B-B-B-B-A-A-A-A-B-B-B-B-

- **ABA-triblock copolymers:** ABA-triblock copolymers have a central **B block** joined by **A blocks** at both ends.

—A—A—A—A—B—B—B—B—B—B—B—B—A—A—A—A—

Graft copolymers

Copolymers that can be prepared by polymerizing a monomer in the presence of a **fully formed polymer of another monomer**.



Classification of Polymers

Chemical Structure

Polymeric structure

- Linear
- Branched
- Crosslinked

Conductivity

Tacticity

Thermal Behaviour

Molecular Force

Methods of Synthesis

Origin

Polymer Structure

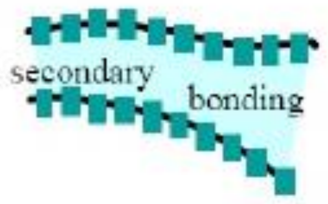
- Linear
- Branched
- Crosslinked

Consists of long string of monomer, attached in a linear manner

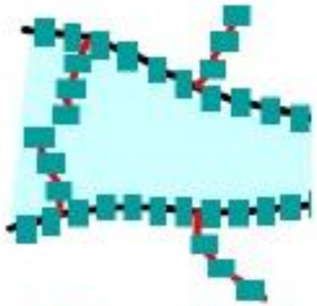
——A——A——A——A——A——A——

Consists of branches at irregular intervals along the polymer chain

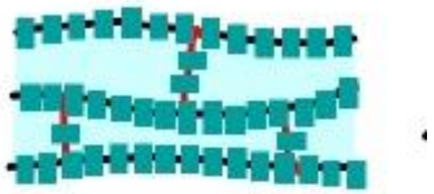
Consists of short side chain that **connect different** polymer chains into a network



Linear



Branched



Cross-linked

Classification of Polymers

Chemical Structure

Polymeric structure

Thermal Behaviour

Thermoplastic

Thermoset

Molecular Force

Methods of Synthesis

Origin

Thermal Behaviour

Thermoplastic

- **Polymers which are easily softened upon heating**
- **Weak intermolecular forces**
- Most **linear** and **slightly branched** polymers are thermoplastic

Ex: Polythene, Polystyrene, Acrylonitrile butadiene styrene (ABS) and polylactic acid (PLA)

Thermosets

- **Polymers which change irreversibly into hard and rigid materials on heating and cannot be reshaped**
- **“sets" irreversibly when heated.**
- Thermosets usually are three-dimensional networked polymers in which there is **a *high degree of cross-linking between polymer chains.***
- **They primarily are used in automobiles, construction, toys, varnishes, boat hulls, and glues**
- **Ex: Bakelite, Malamine Formaldehyde, Epoxy Resins**

Thermoplastic	Thermoset

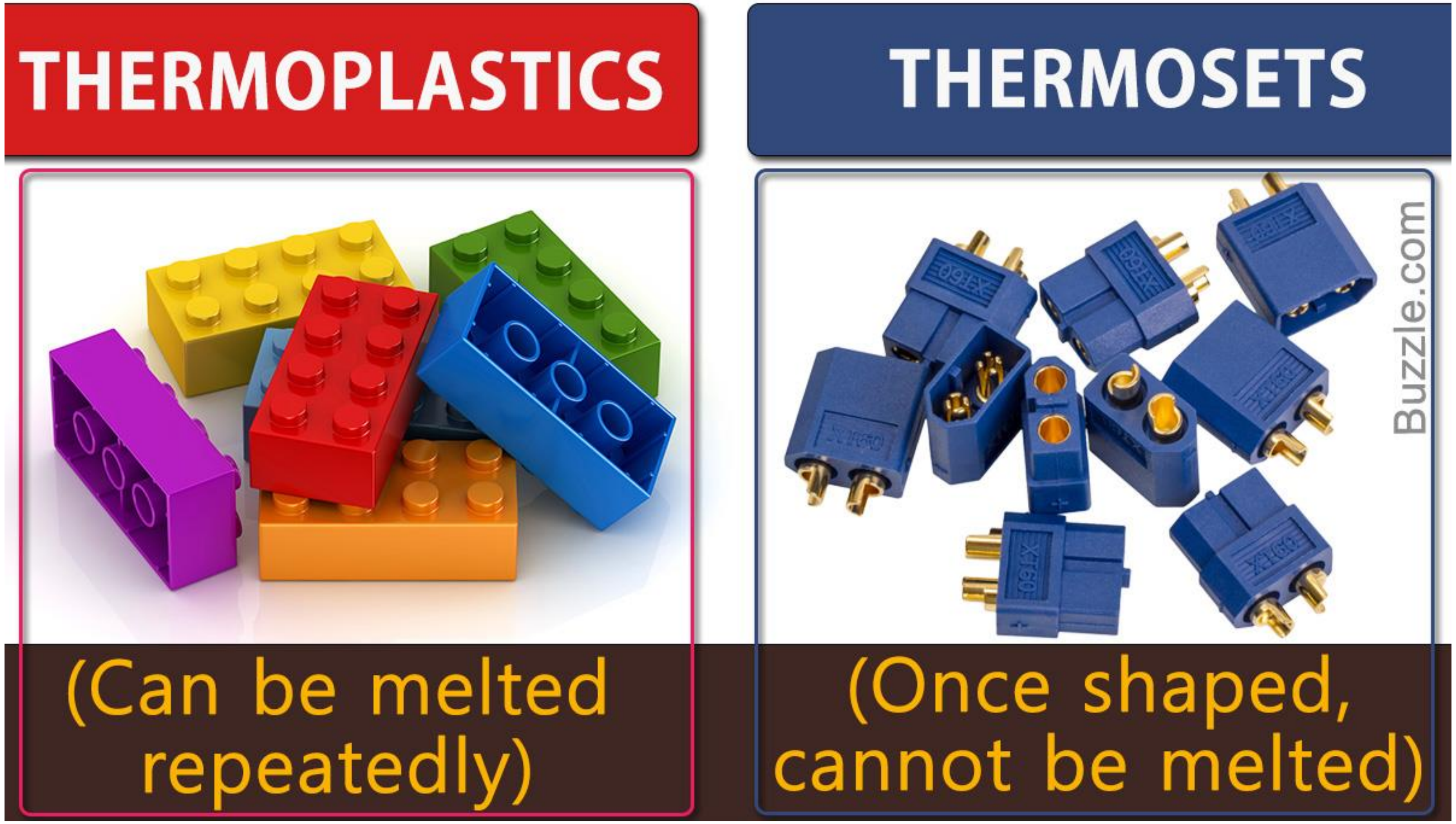
Thermoplastic

Molecules in a thermoplastic are held together by relatively weak intermolecular forces so that the material *softens when exposed to heat and then returns to its original condition when cooled*

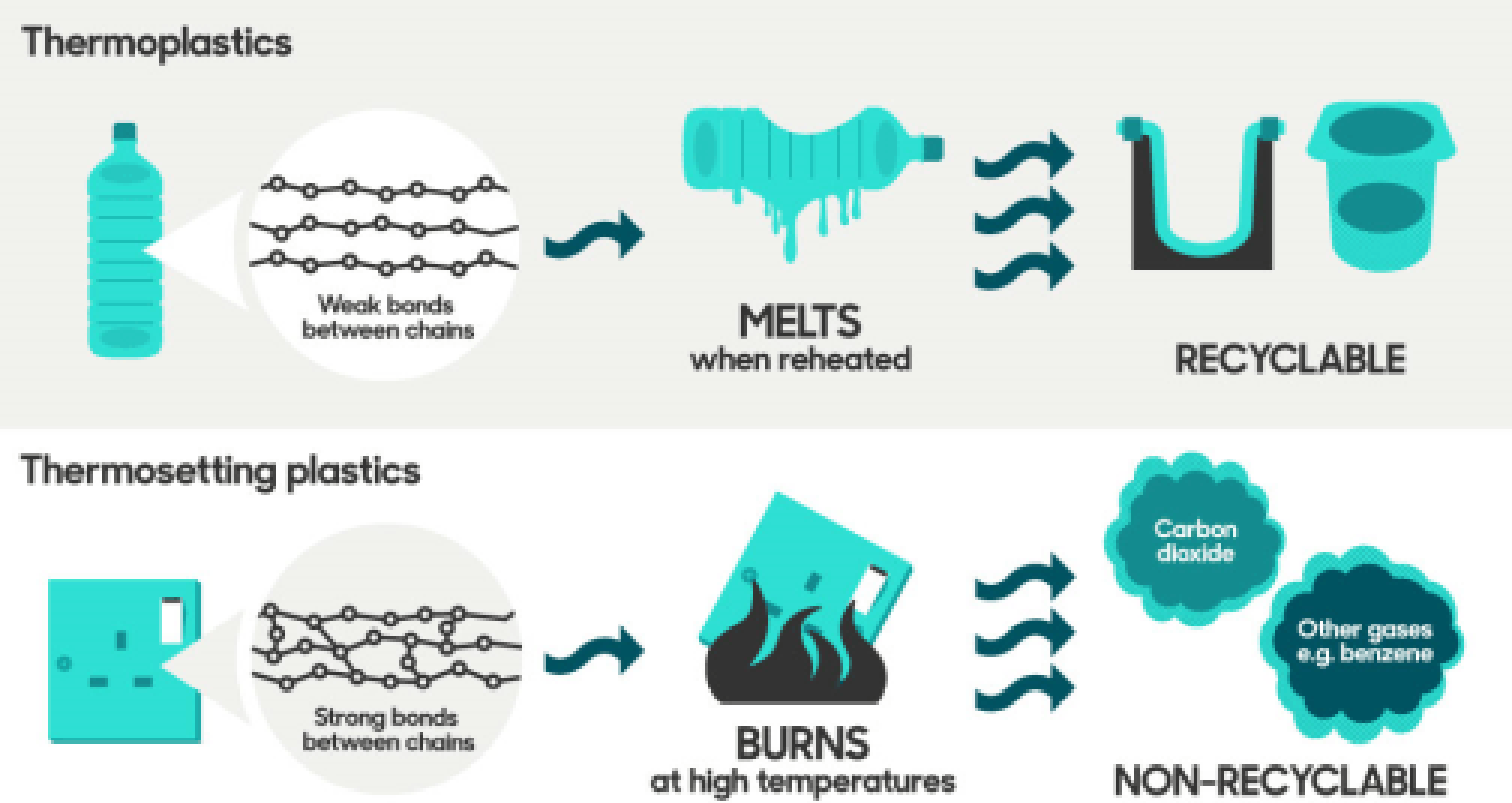
Thermoset

- Thermosets cannot be **reshaped** by heating.
- Thermosets usually are three-dimensional networked polymers in which there is **a *high degree of cross-linking between polymer chains*.**
- **The cross-linking restricts the motion of the chains and leads to a rigid material.**
- **Thermosets are strong and durable.**
- **They primarily are used in automobiles, construction, toys, varnishes, boat hulls, and glues**

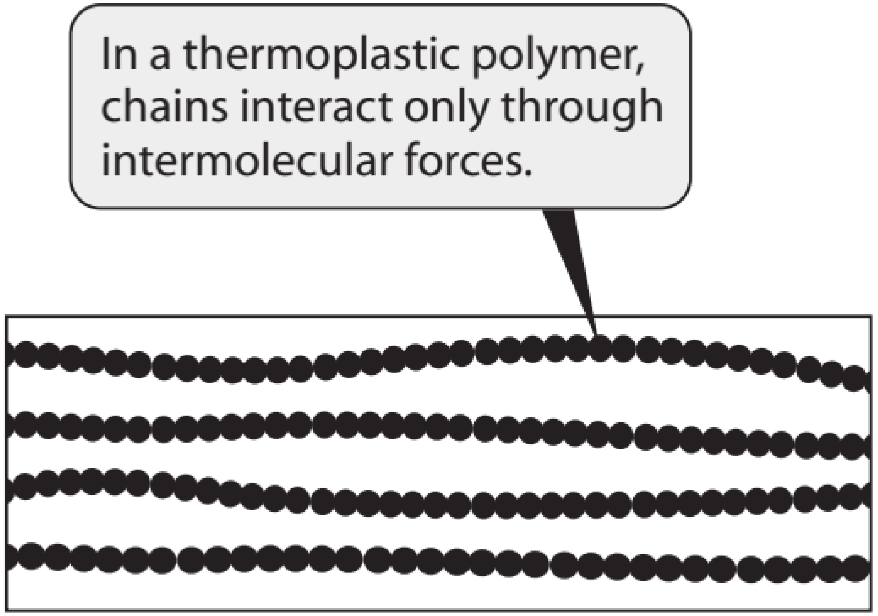
Thermoplastics and Thermosets

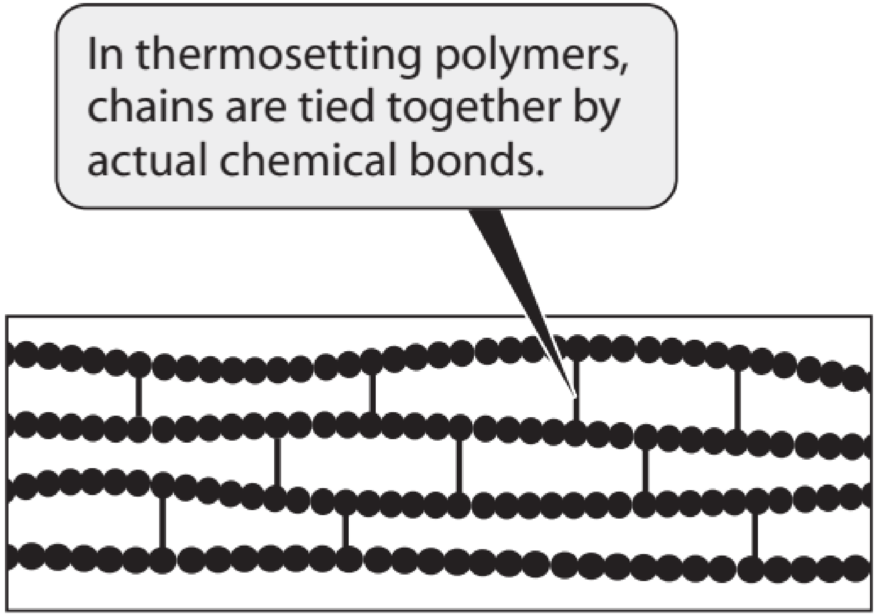


Thermoplastics and Thermosets

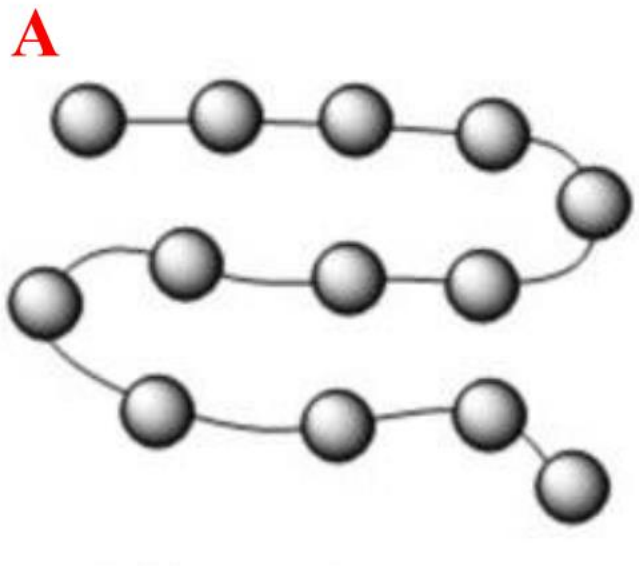


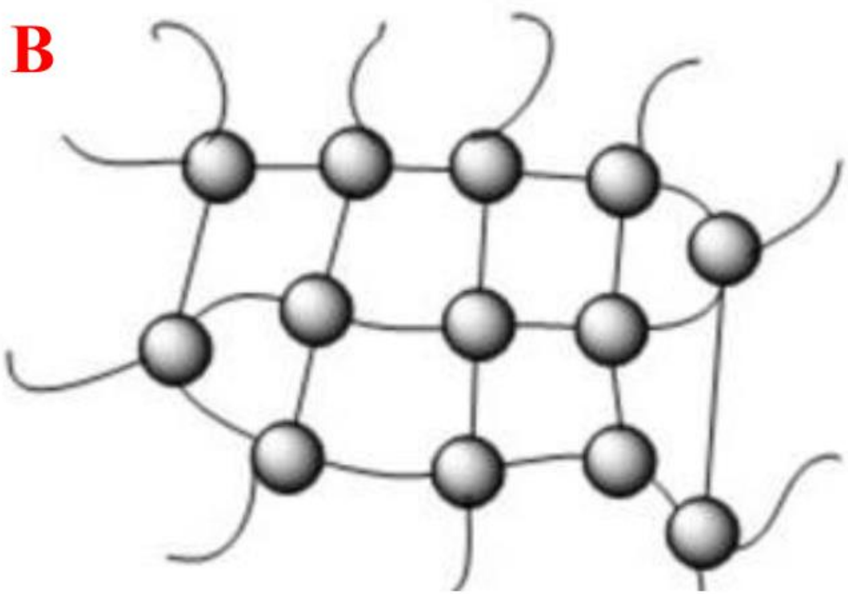
Thermoplastics and Thermoset





In the following picture two polymer structures are represented by linking various units of monomer (grey balls). Find out which of them are thermoset and thermoplastic?





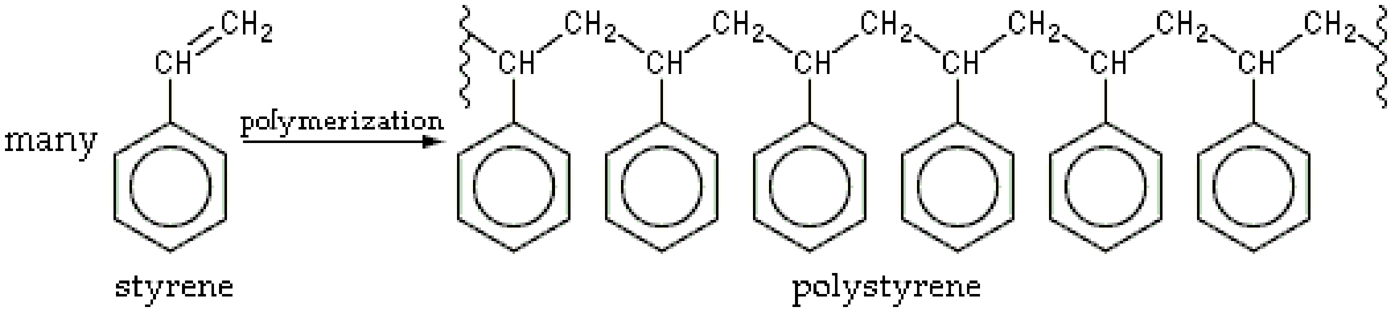
Thermoplastic

Thermoset

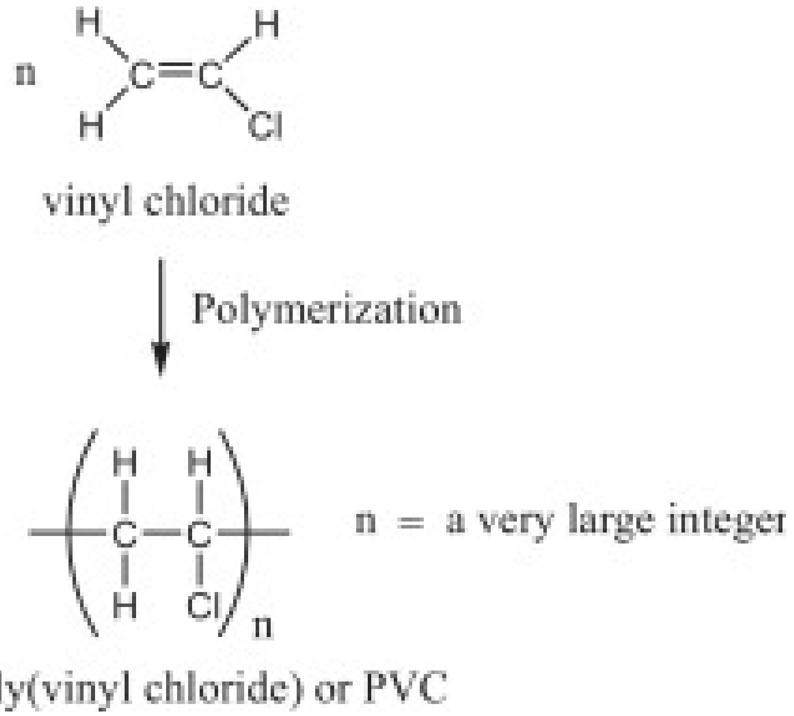
Polymers-Classification

Thermoplastics

Polystyrene



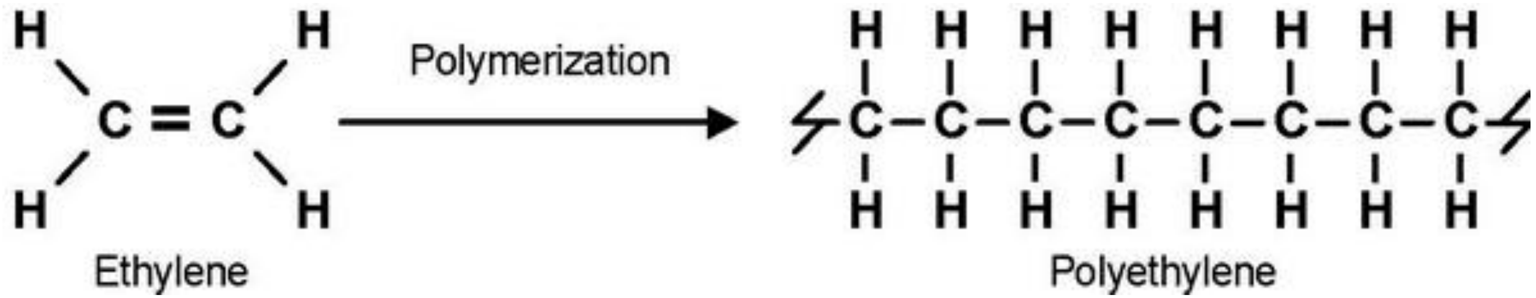
Polyvinyl chloride



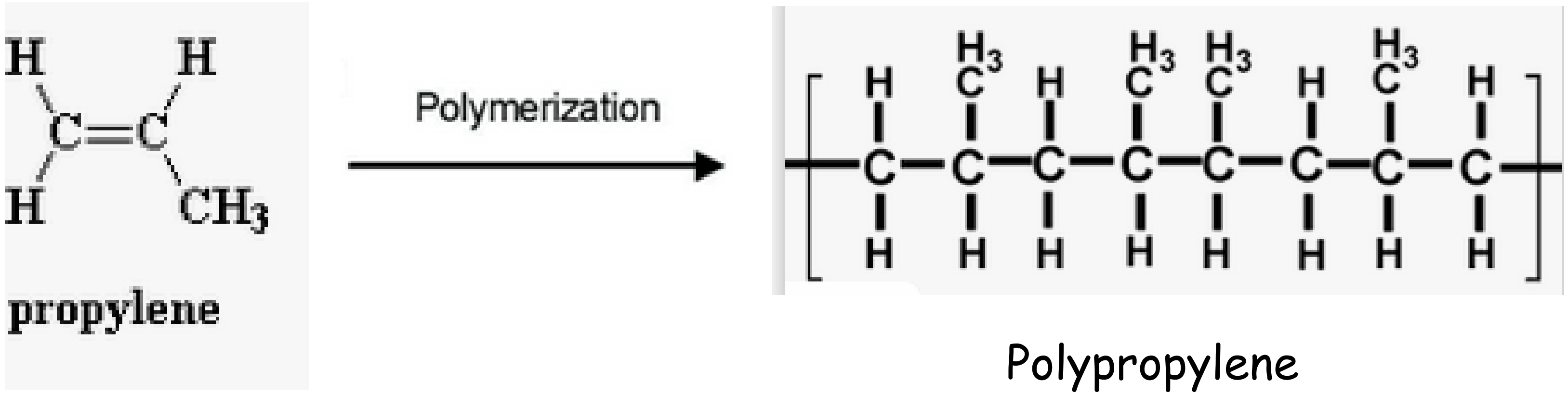
Polymers-Classification

Thermoplastics

Polyethylene



Polypropylene



Polypropylene

Polymers-Classification

Thermoset (Thermosetting)

Some polymers undergo chemical changes and cross-linking on heating and become permanently hard, rigid and infusible on cooling. These are called thermosetting polymers.

Once formed, these cross-linked networks resist heat softening, mechanical deformation, and solvent attack, but cannot be thermally processed.

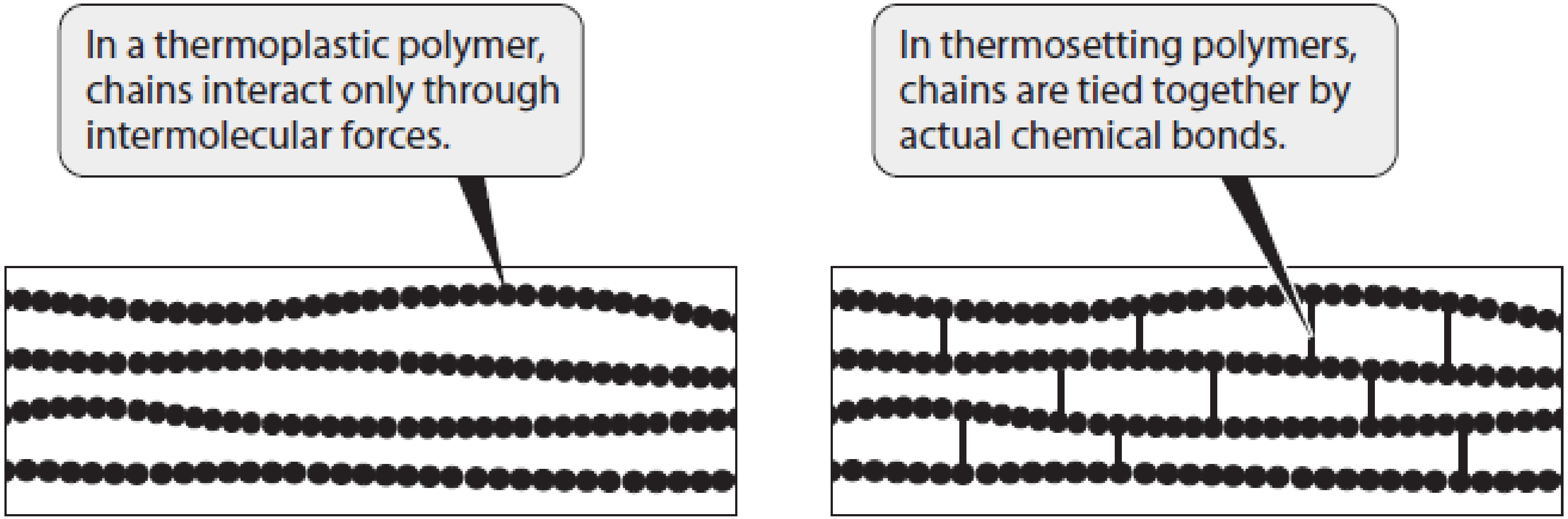
Polymers-Classification

Thermoset (Thermosetting)-Applications

Such properties make thermosets suitable materials for composites, coatings, and adhesive applications.

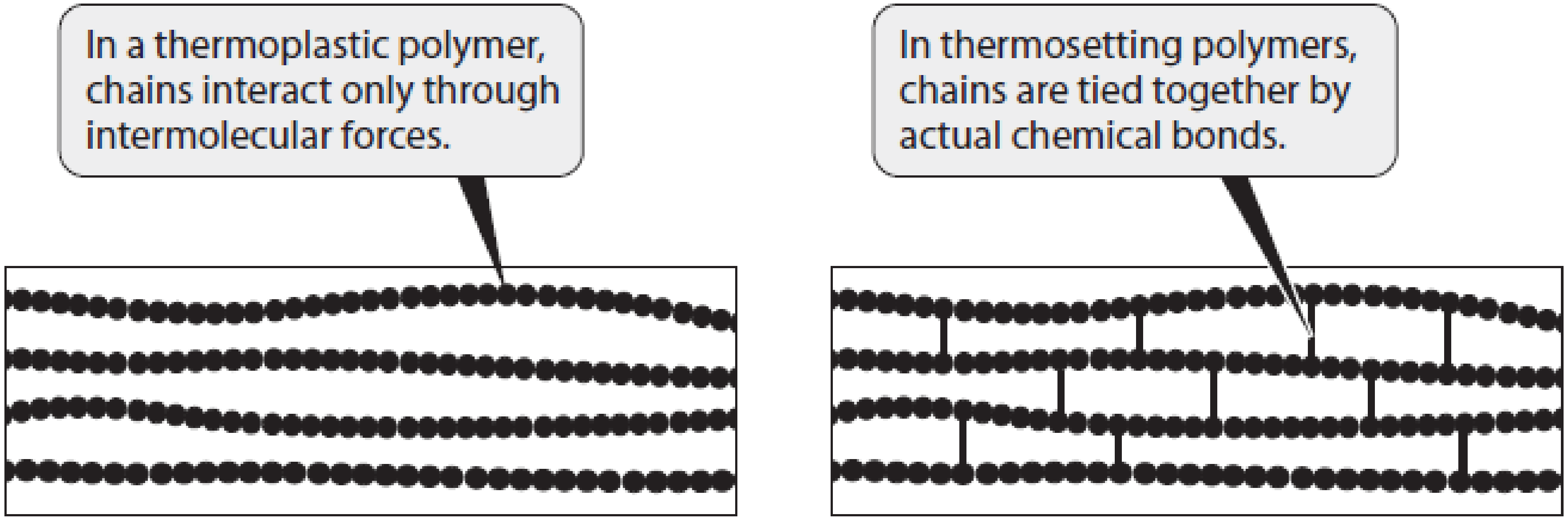
Principal examples of thermosets include epoxy, phenol-formaldehyde resins, and unsaturated polyesters that are used in the manufacture of glass-reinforced composites such as Fiberglass

Additional information on thermoplastic and thermoset



The initial heating and setting of a thermosetting polymer produces a number of links between sites on the carbon backbone of different molecular chains. These links are referred to as crosslinks because they cross between and link individual molecular strands of the polymer.

Additional information on thermoplastic and thermoset

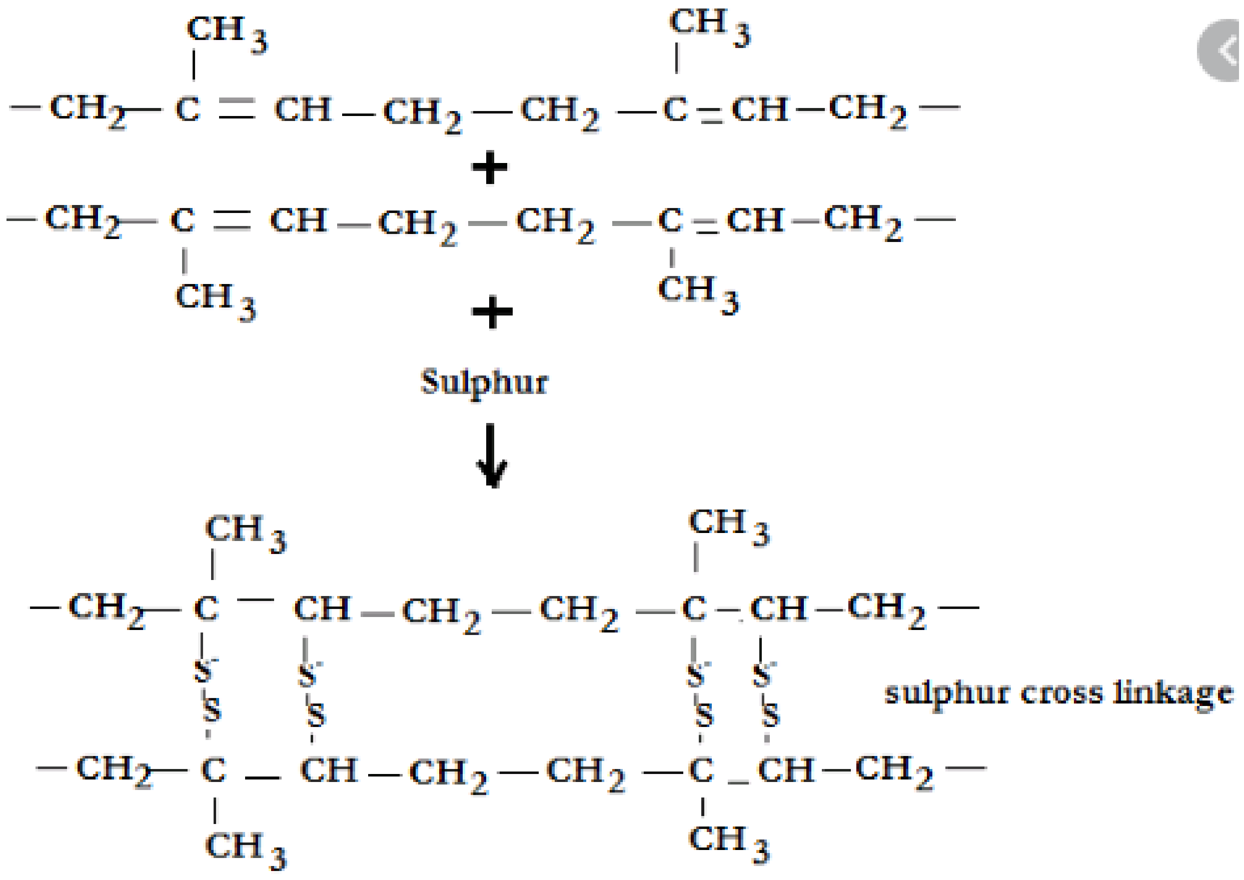


Chemically, these cross-links are additional covalent bonds that join the polymer chains to one another. Like most covalent bonds, they are strong enough that they do not readily fail upon heating. So the cross-linked polymer keeps its shape.

Example: vulcanization

Additional information on thermoplastic and thermoset

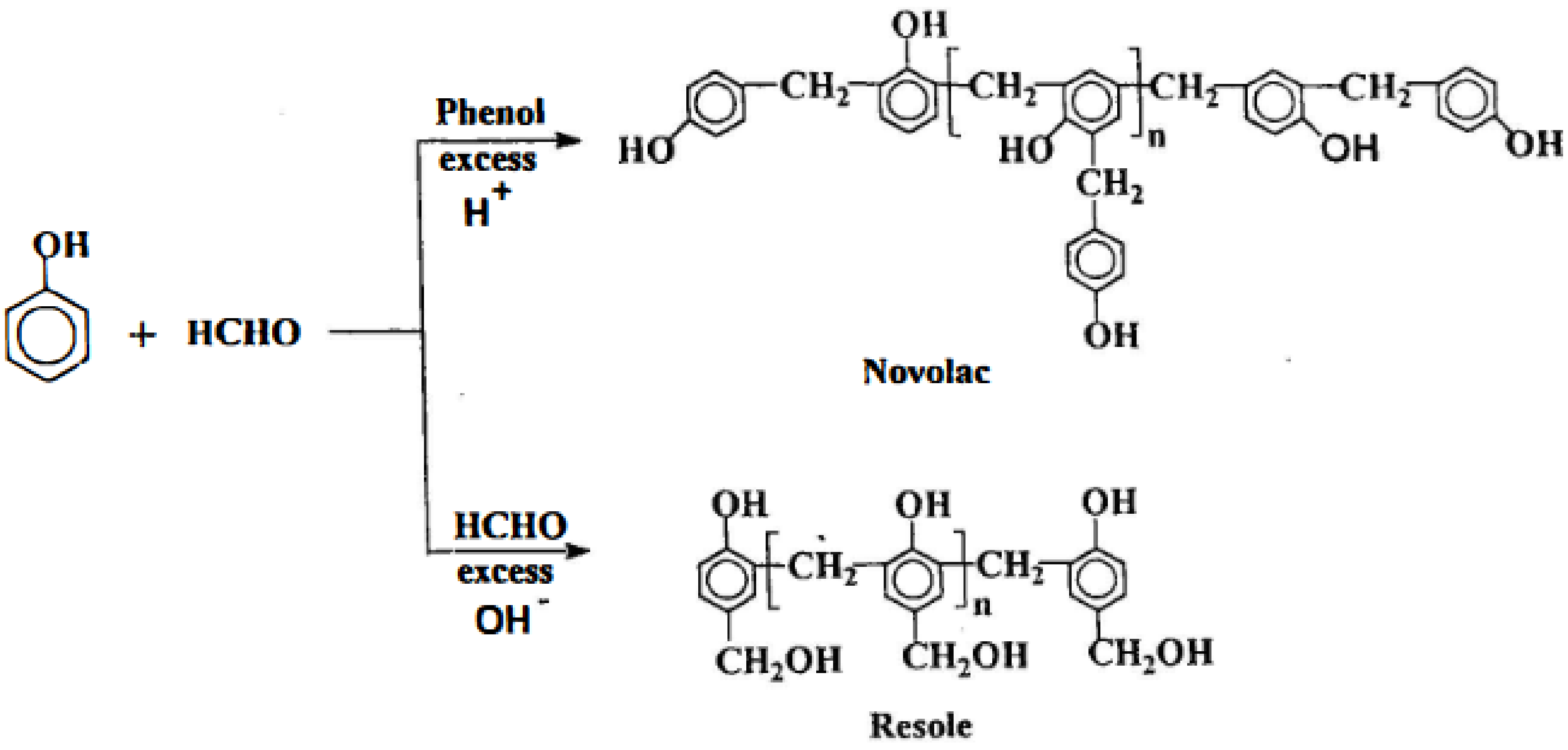
Example: vulcanization



Phenolic resins: Bakelite

- Bakelite was the first plastic made from synthetic components.
- It is a thermosetting phenol formaldehyde resin, formed from a condensation reaction of phenol with formaldehyde.

Synthesis of Bakelite



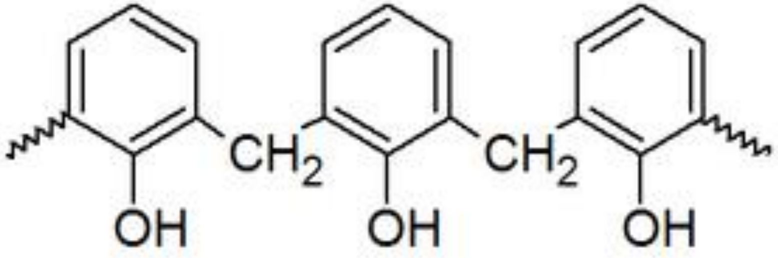
A-stage $\xrightarrow{\text{heating}}$ B-stage $\xrightarrow{\text{heating}}$ C-stage

first stage of reaction, no cross-linking

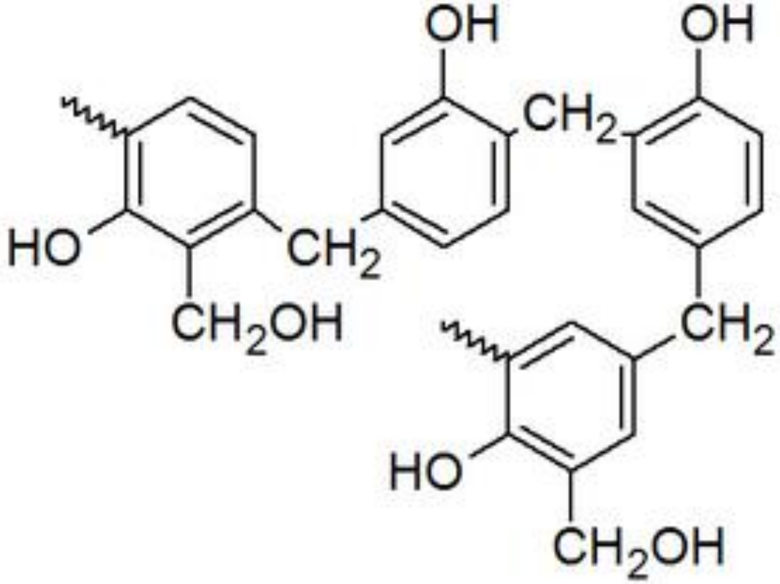
second stage of reaction, partially cross-linking

third stage of reaction, more cross-linking

Synthesis of Bakelite



Novolac

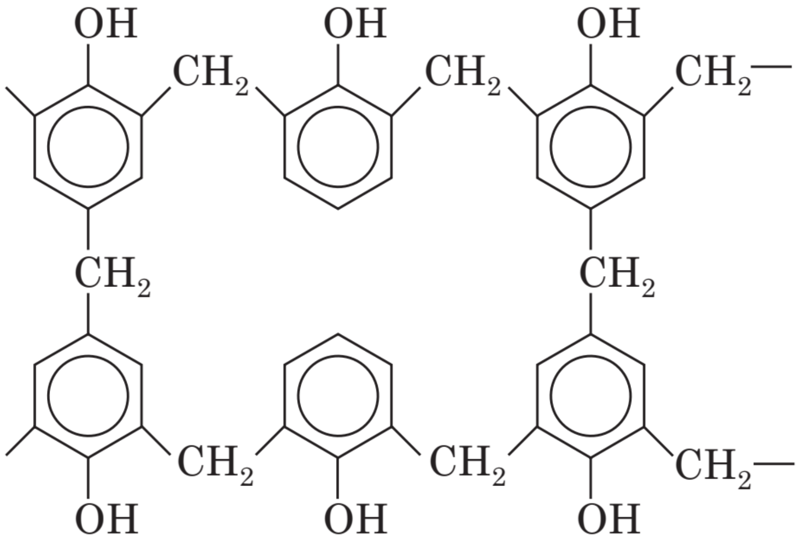


Resole

Synthesis of Bakelite

Bakelite is crosslinked polymer

$$\text{Novolac} \xrightarrow{\text{cross-linking}} \text{Bakelite}$$



bakelite

C-Stage Resin

Bakelite: Properties and Uses

Properties. These phenolic resins are rigid, hard, water resistant. They are resistant to non-oxidising acids, organic solvents but are susceptible to alkalies. Solubilities and melting point of the resin gradually change with rise of molecular weight. These resins possess electrical insulating properties.

Uses. It can be widely used as metal substitute where high tensile strength is not necessary. As an inert material it can substitute for glass. It can be used for making insulator parts like switches, plugs, heater handles. It can be moulded to cabinets for TVs and radio and telephone parts. It is used as adhesive also used in paints and varnishes, as cation exchanger resin for water softening, in paper industry as propeller shafts.

Classification of Polymers

Chemical Structure

Polymeric structure

Conductivity

Tacticity

Thermal Behaviour

Molecular Force

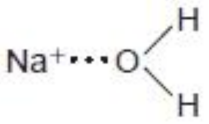
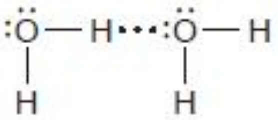
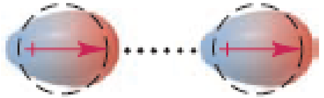
Elastomers

Fibers

Methods of Synthesis

Origin

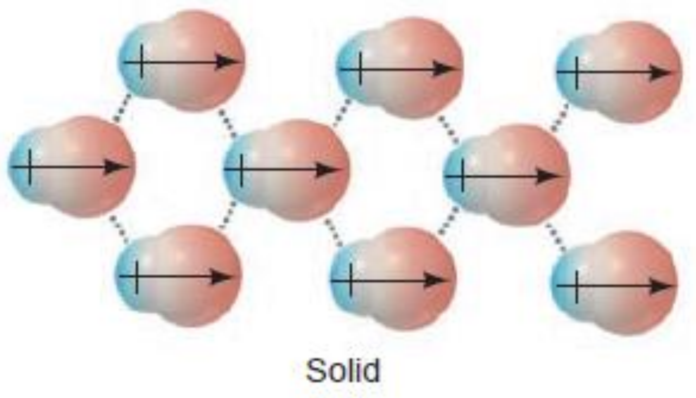
Intermolecular forces

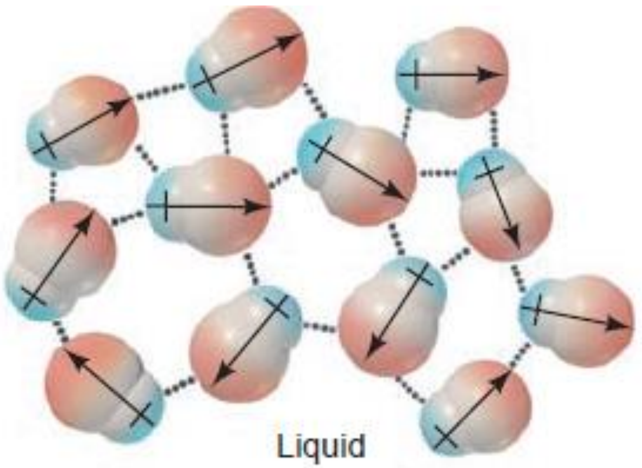
Force	Model	Basis of Attraction	Energy (kJ/mol)	Example
Nonbonding (Intermolecular)				
Ion-dipole		Ion charge–dipole charge	40–600	
H bond	δ– δ+ δ– —A—H·······:B—	Polar bond to H–dipole charge (high EN of N, O, F)	10–40	
Dipole-dipole		Dipole charges	5–25	I—Cl···I—Cl
Dispersion (London)		Polarizable e^- clouds	0.05–40	F—F···F—F

Intermolecular forces

- **Dipole-Dipole Forces**

When polar molecules lie near one another, as in liquids and solids, their partial charges act as tiny electric fields that orient them and give rise to **dipole-dipole forces**.



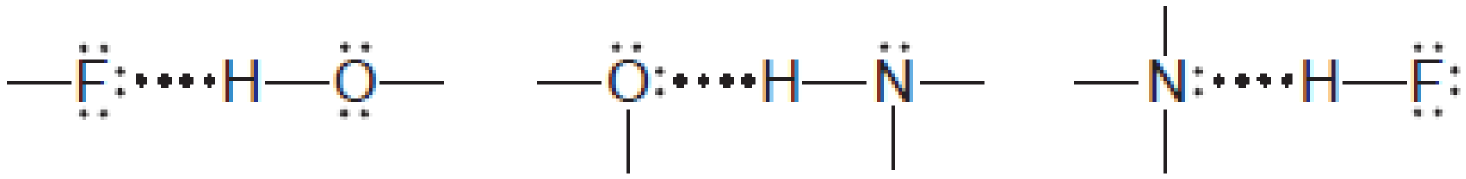


Intermolecular forces

- **The Hydrogen Bond**

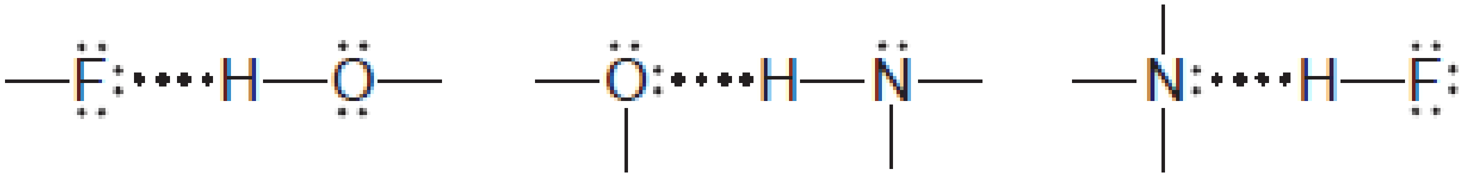
Hydrogen bond forms between molecules that have an H atom bonded to a small, highly electronegative atom with lone electron pairs.

The most important atoms that fit this description are **N**, **O**, and **F**.



Intermolecular forces

- **The Hydrogen Bond**



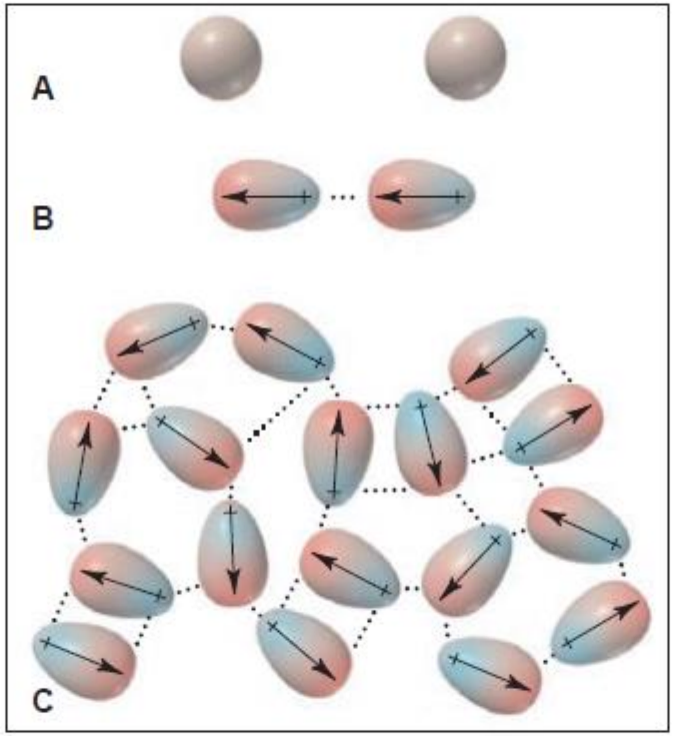
- The H—N, H—O, and H—F bonds are very polar, so electron density is withdrawn from H. As a result, the partially positive H of one molecule is attracted to the partially negative lone pair on the N, O, or F of another molecule

- Hydrogen bonding is a **strong type of dipole-dipole force**

Intermolecular forces

Dispersion forces

Dispersion forces are caused by momentary oscillations of electron charge in atoms and, therefore, are present between all particles (atoms, ions, and molecules).



Separated **Ar** atoms are nonpolar

An instantaneous dipole in one atom induces a dipole in its neighbour. These partial charges attract the atoms together.

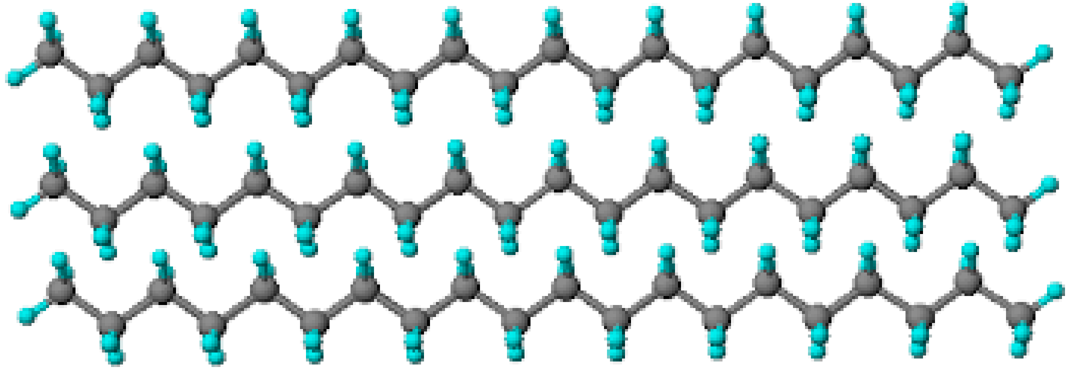
This process takes place among atoms throughout the sample.

nonpolar molecules show dispersion forces.

Intermolecular force

Weak force of attraction between the molecules

Inter polymer chain or intermolecular forces

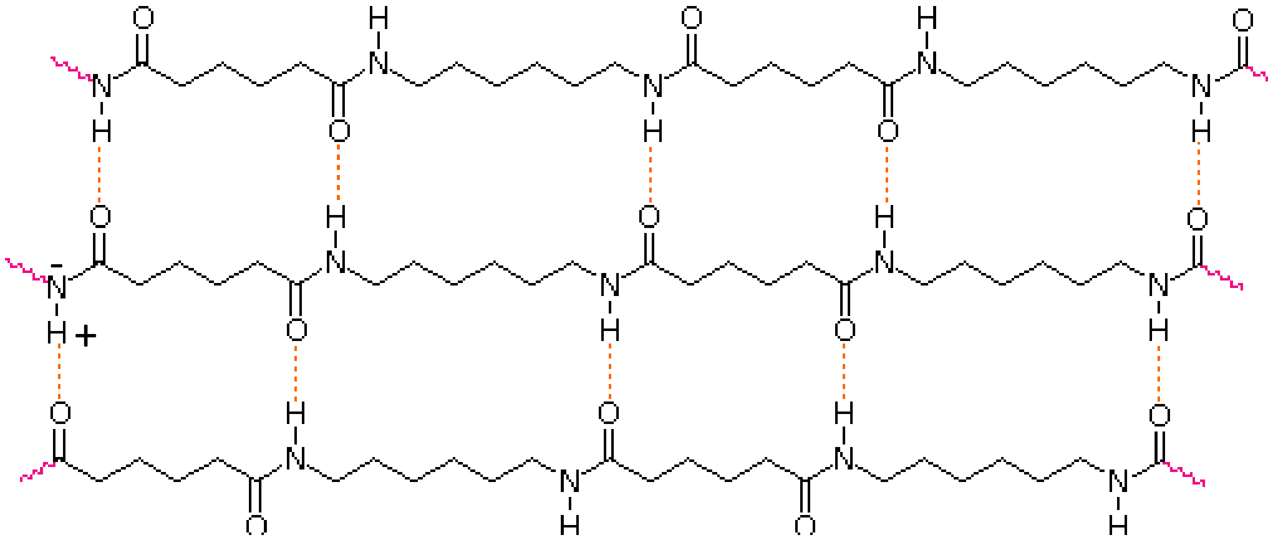


Polyethylene – Long chain hydrocarbon

Instantaneous dipole –induced dipole
Or dispersion interaction

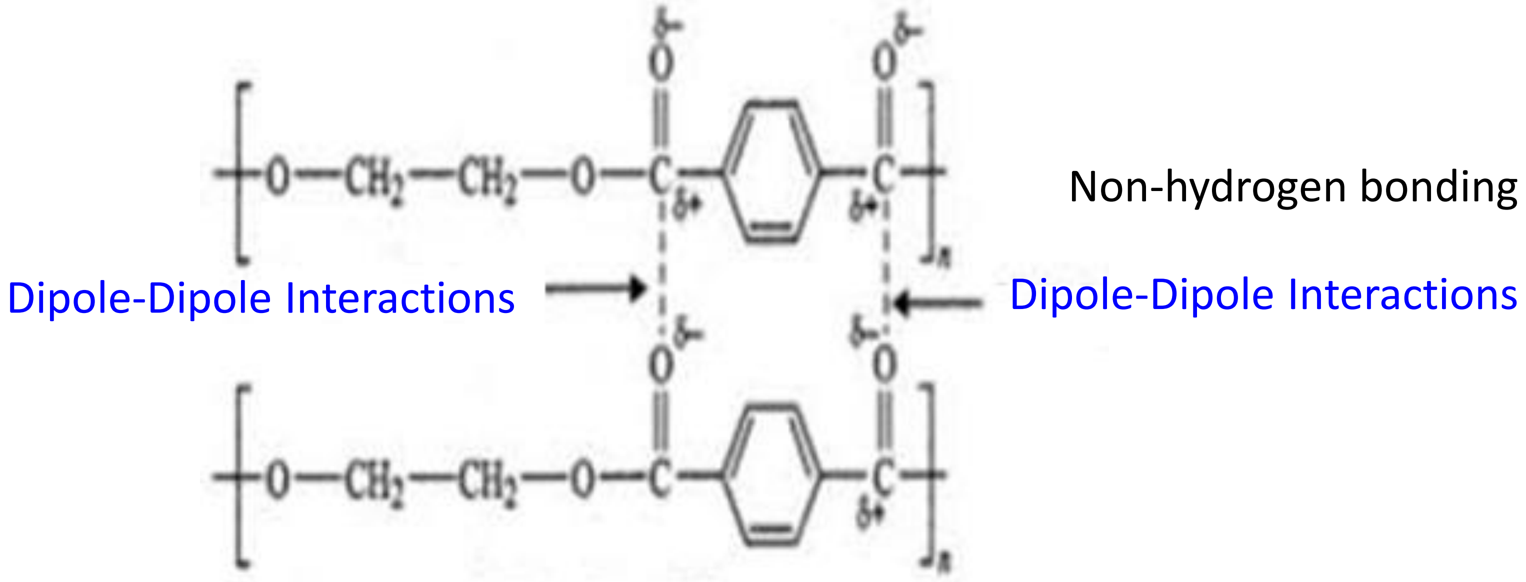
Polyamides
Hydrogen bonding

Permanent dipole



Polyester

Dipole-Dipole Interactions



The extent of slipping away of the chains from one another depends on the strength of the interchain force.

Strength of intermolecular force: Hydrogen bond > Dipole-Dipole > Dispersion force

"Tendency of chains to slip away is in reverse order"

Hydrogen bonded chains have less tendency to slip away than that of dipole-dipole bonded systems.

"Therefore hydrogen bonded systems are tough fibers with large stiffness and elastic modulus". (Example cellulose)

Classification of Polymers

Chemical Structure

Polymeric structure

Conductivity

Tacticity

Thermal Behaviour

Molecular Force

- Elastomers
- Fibers

Methods of Synthesis

Origin

Intermolecular Forces

Elastomers

- Polymers that can be easily stretched by applying small stress

- Absence of stiffening groups in the polymer Backbone

- Presence of bulky pendant groups

- Absence of strong intermolecular forces.

Ex: Rubber

Fibers

- Polymers which have strong intermolecular forces like 'Hydrogen bonding' or dipole-dipole interaction between polymer chain

- Presence of stiffening groups
- Presence of 'Hydrogen bond'

- Absence of branching or irregularly spaced pendant groups

Ex: Nylon 6,6 Dacron

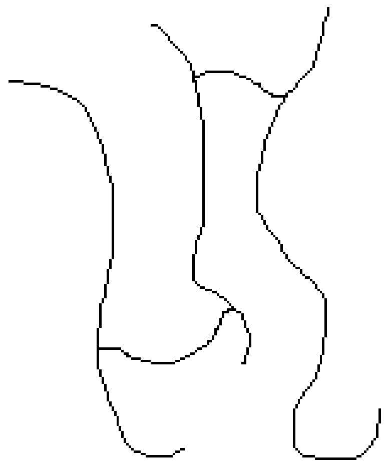
Elastomers

"Elastomers are rubbery polymers that can be stretched easily and which rapidly return to their original dimensions when the applied stress is released"

It can be done reversibly for several times.

Elastomers are **have weakest intermolecular forces** between the chains.

A few cross-links are introduced between the chains to retract the polymer to active it's original dimension.



In order for polymer to be elastic, it must be above its glass transition temperature, Tg, and have a low degree of crystallinity.
Rubber bands and other elastics are made of elastomers.

Classification of Polymers

Chemical Structure

Polymeric structure

Conductivity

Thermal Behaviour

Molecular Force

Backbone

Crystallinity

- Crystalline polymer
- Amorphous Polymer

Origin

Crystallinity

- Crystalline
  - Ordered

- Amorphous
  - Random

- Semi-crystalline
  - Consists of both

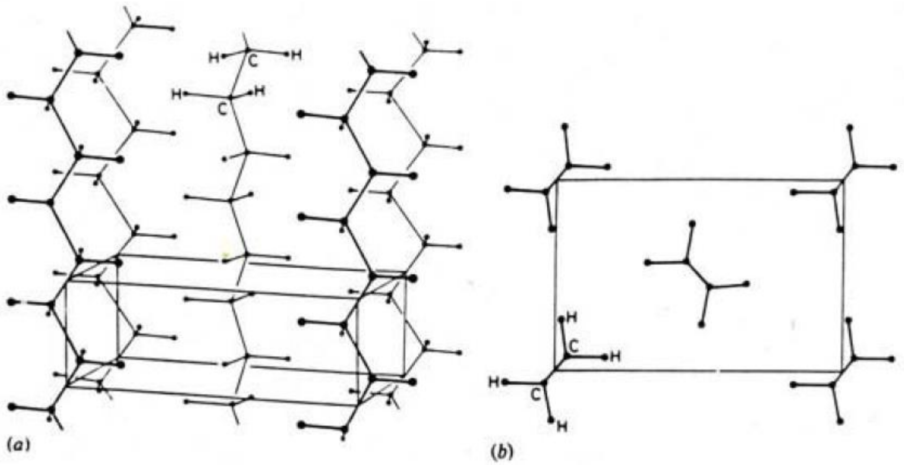
Chemical structure- Crystalline Vs Amorphous

Crystallinity:

The requirement for a polymer to crystallize is regularity of molecular structure with greater symmetry, such that polymer chains can pack up closely in an ordered structure.

The degree of crystallinity of the polymer depends on its structure.

Examples linear polyethylene and isotactic polypropylene



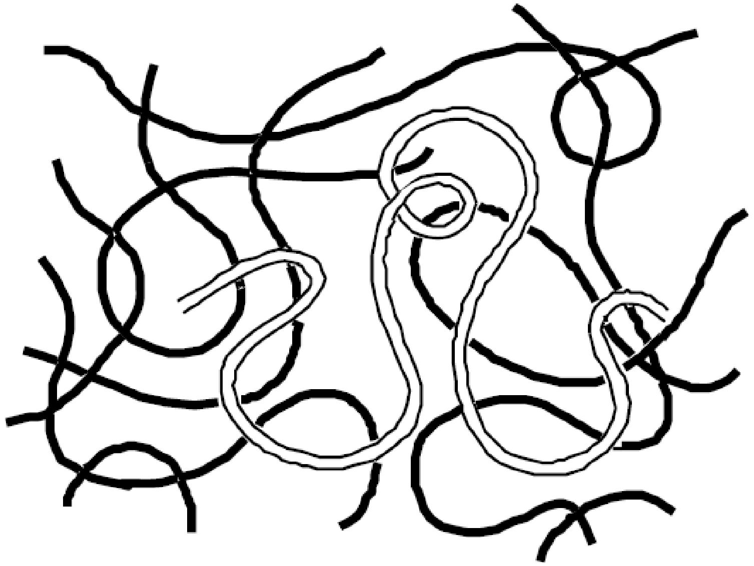
Crystalline polyethylene: orthorhombic

In crystalline polymers, the atomic arrangements follows a repeating pattern in long range order.

Chemical structure -Crystalline Vs Amorphous

Amorphos polymers

In many cases, individual chains are randomly coiled and intertwined with no molecular order or structure. Such a physical state is termed amorphous.

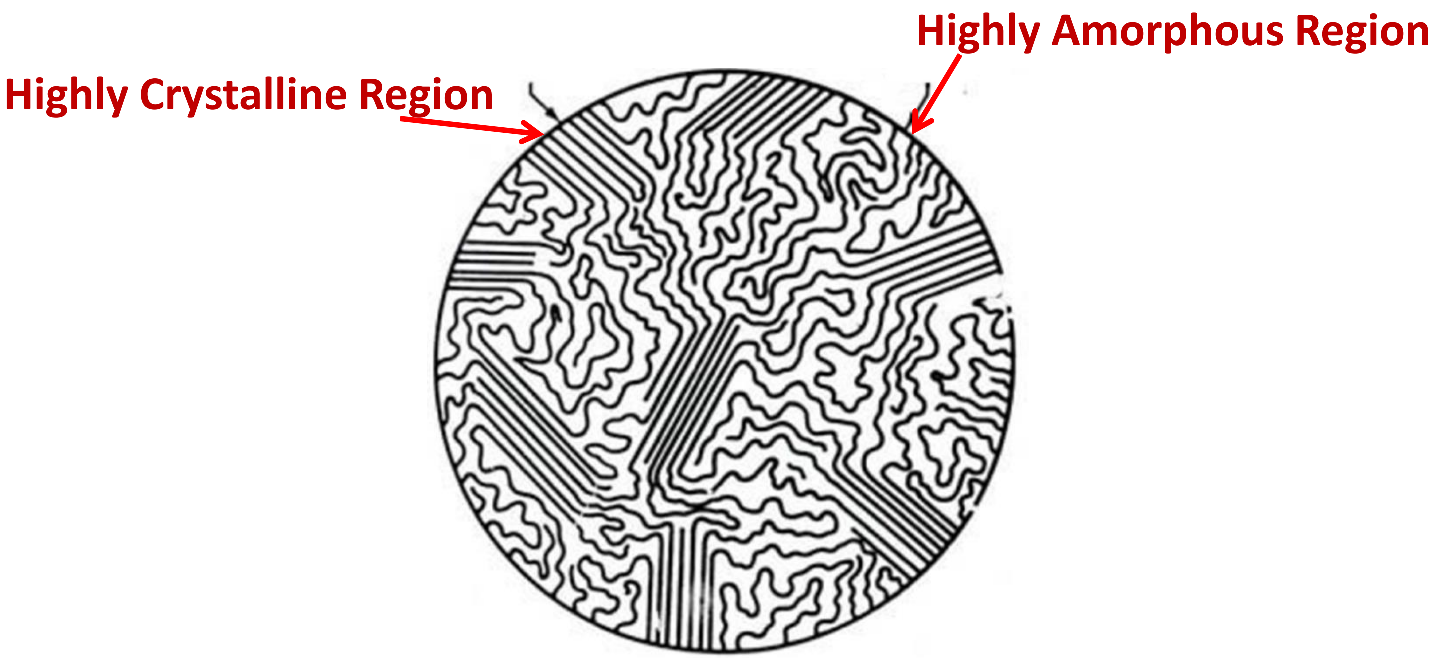


Example: atactic polystyrene

Semicrystallinity:

- Although single crystals of some polymers such as polyethylene can be grown under laboratory conditions, no bulk polymer is completely crystalline.

- In the case of semicrystalline polymers, regular crystalline units are linked by unoriented, random-conformation chains that constitute amorphous regions.

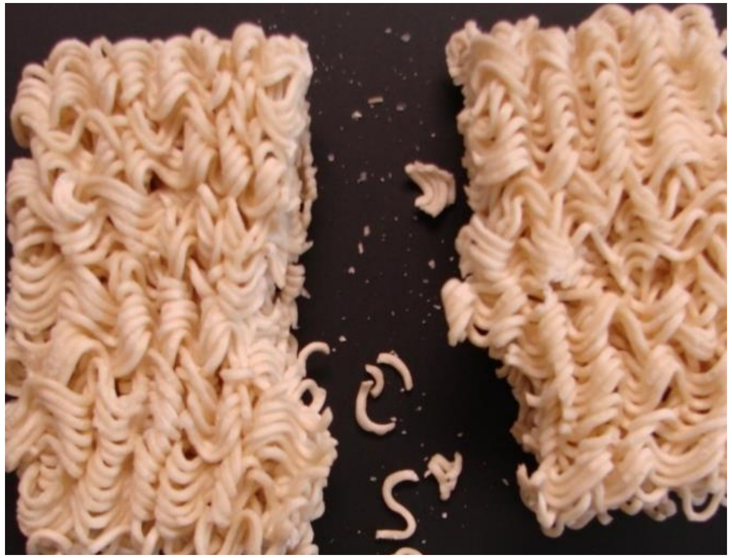


Examples

poly(vinyl chloride)

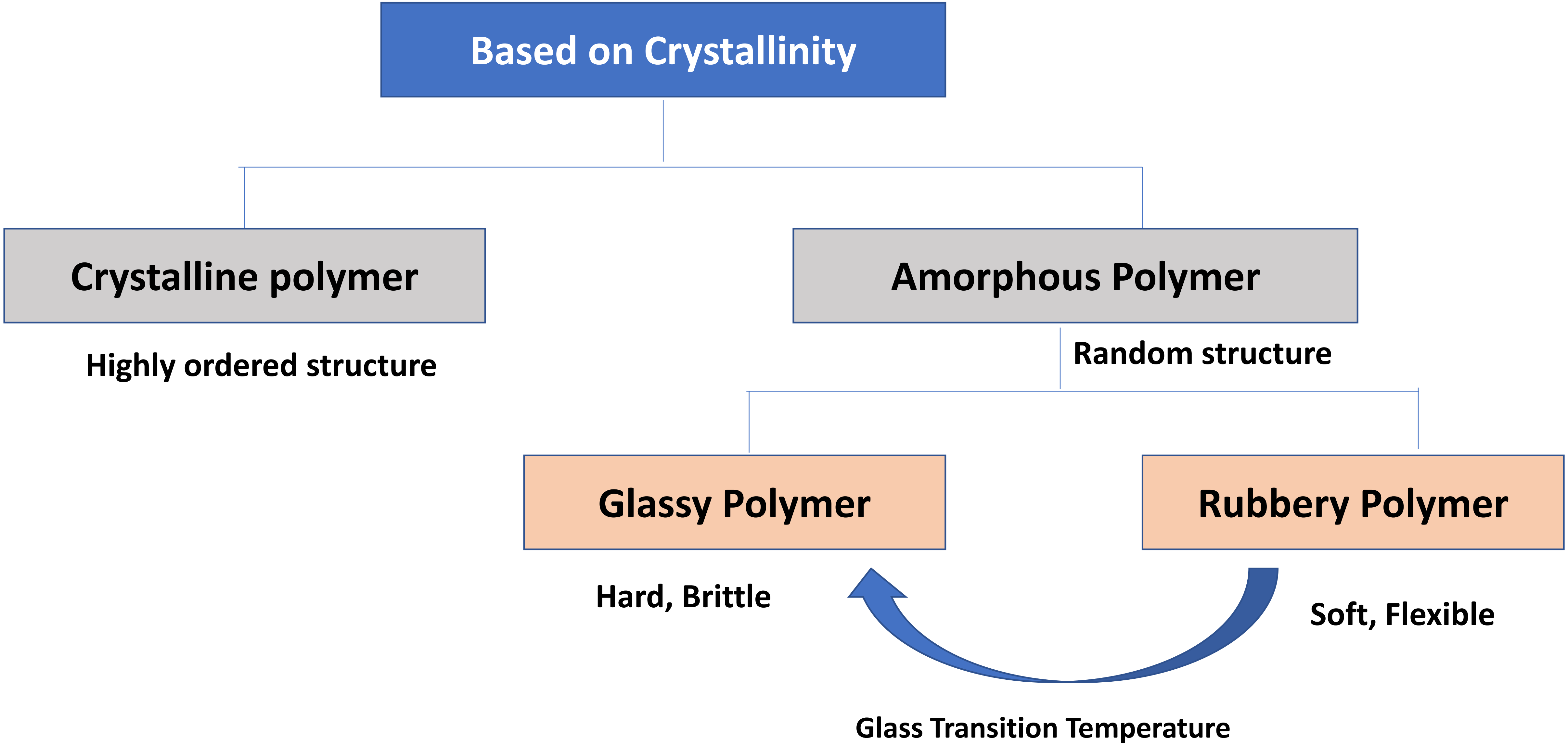
Concept Examples : Morphology

Think of Semi-crystalline materials like **maggi noodles**. When in their solid state, they have a compact ordered arrangement



Amorphous materials are like **cooked noodles** in that there is a random arrangement

Glass Transition Temperature (T_g)

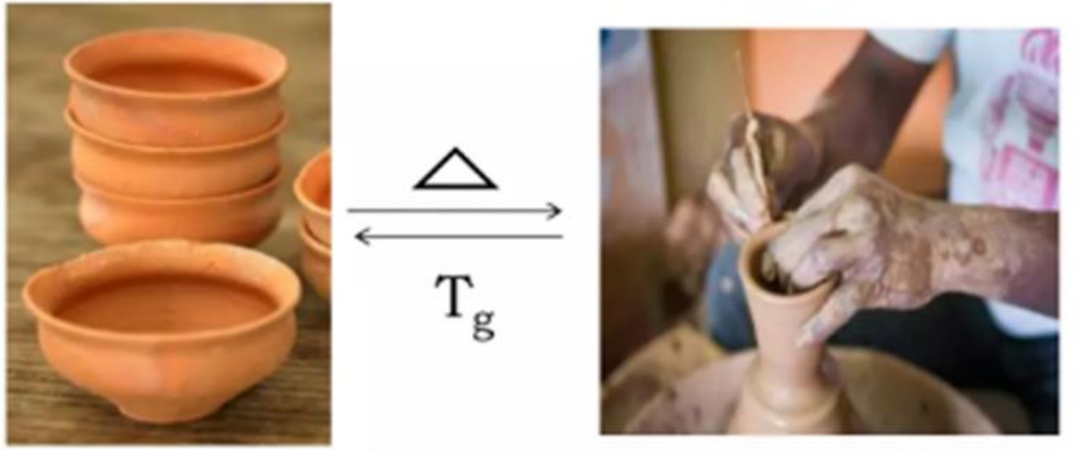


Concept of T_g – in Chewing Gum

- At body temperature the gum is **soft and pliable**, which is characteristic of an amorphous solid in the **rubbery state**.
- If you put a cold drink in your mouth or hold an ice cube on the gum, it becomes **hard and rigid**.
- The glass transition temperature of the gum is somewhere between 0 °C and 37 °C.

Glass Transition Temperature (Tg)

- Glass transition temperature is a temperature at which the polymer experiences the transition from the **glassy state** to the **rubbery state**.



Glassy State
Hard & Brittle

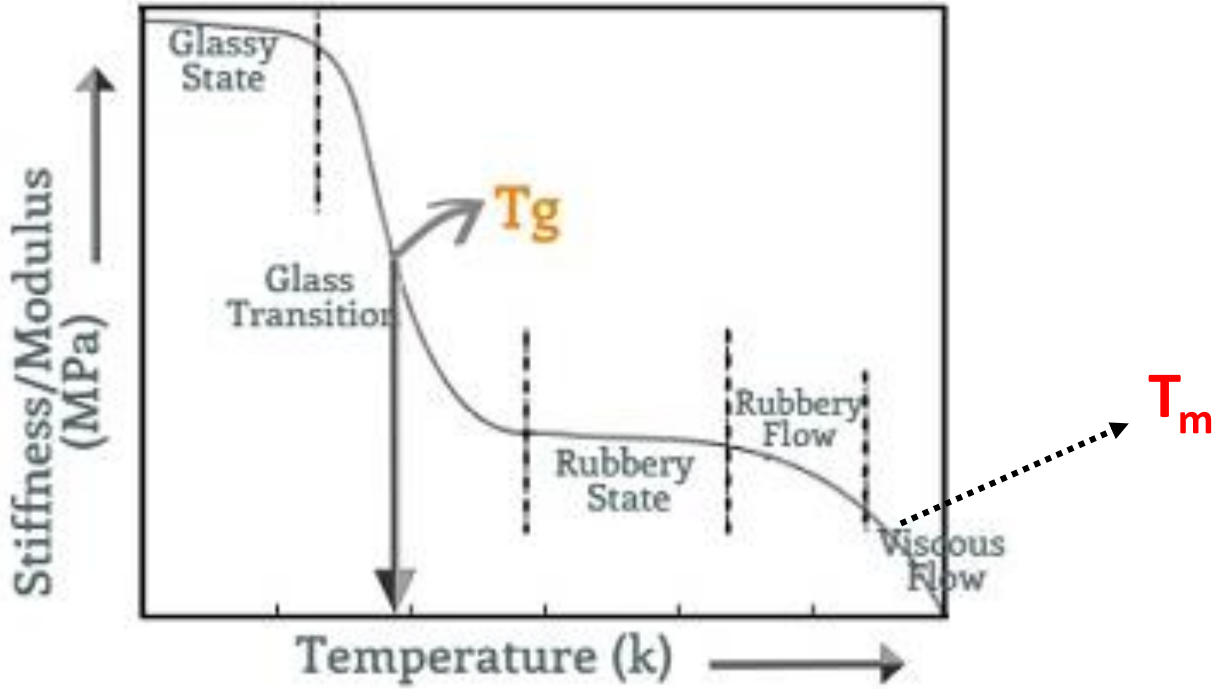
Rubbery State
Soft & Flexible

Glassy State:

- Hard & Brittle state of material
- Consist of short-range vibrational and rotational motions of atoms in polymer chain

Rubbery State:

- Soft & Flexible
- Long-range rotational motion of polymer chain segments



- Below T_g : Polymers are brittle like glass, due to lack of mobility
- Above T_g: Polymers are soft and flexible like rubber due to some mobility

- ***Crystalline-melting temperature*** (T_m) is a temperature at which crystalline substance becomes amorphous. *(Solid to liquid)*

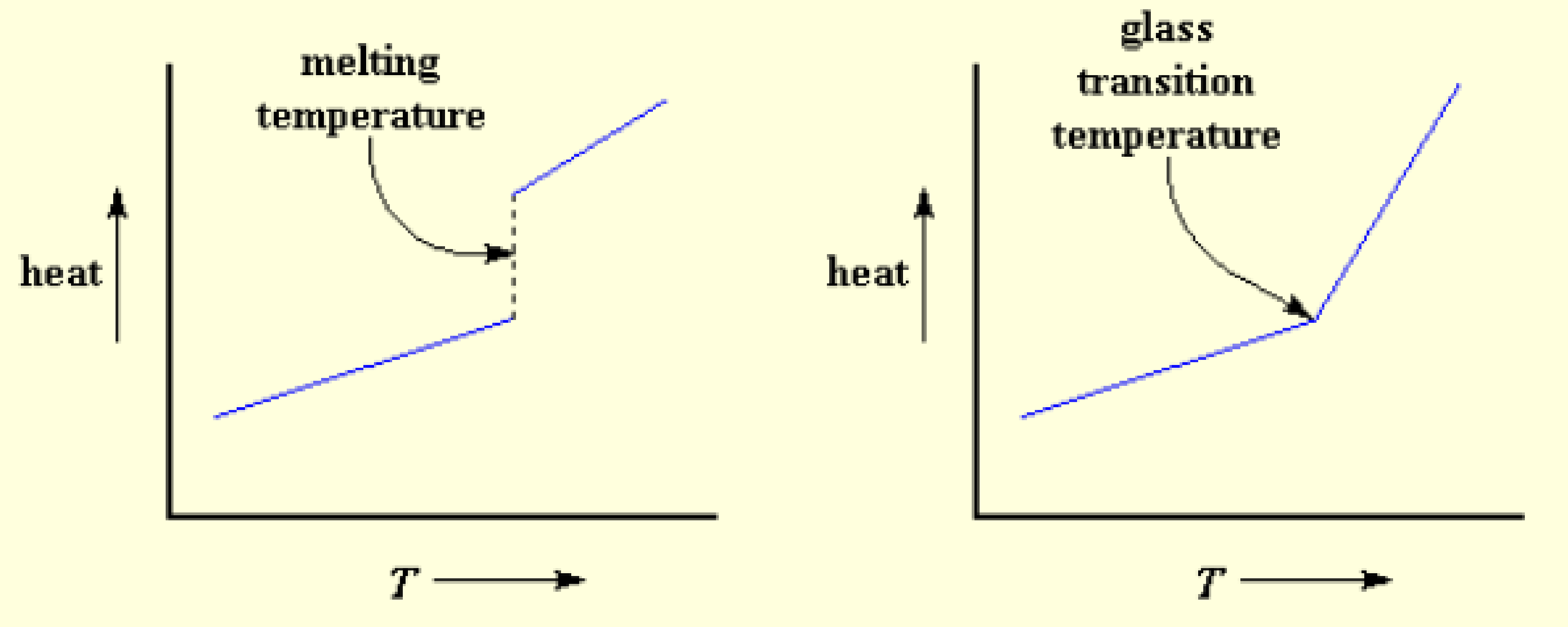
Glass Transition Temperature (Tg)

- Glass transition temperature is a temperature at which the polymer experiences the transition from the **glassy state** to the **rubbery state**.

- Some polymers are used above their glass transition temperature, and some are used below

- **Hard plastics** like polystyrene and poly methyl methacrylate are **used below** their glass transition temperature; that is in their glassy state. Their Tg's are well above room temperature

- **Elastomers** like polyisoprene and polyisobutylene are used **above their Tg's**, that is in the rubbery state, where they are soft & flexible



A heat vs. temperature plot for an crystalline polymer, on the left; and a amorphous polymer on the right.

Comparison of T_g with Melting

Crystalline-melting temperature or T_m is a temperature at which crystalline–amorphous transition occurs. The glass transition is NOT the same as melting.

- **Glass Transition**
  - Property of the amorphous region
  - Below T_g: Disordered amorphous solid with immobile molecules
  - Above T_g: Disordered amorphous solid in which portions of molecules can wiggle around
- **Melting**
  - Property of the crystalline region
  - Below T_m: Ordered crystalline solid
  - Above T_m: Disordered melt

Factors effecting T_g

(a) Molecular Weight

- T_g is directly **proportional** to molecular weight

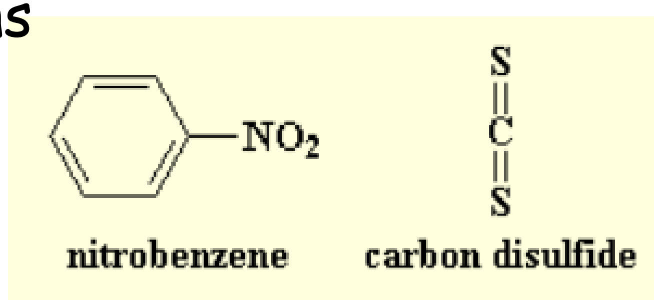
(b) Cross linking

- Tg is directly **proportional** to cross links
- Cross links restricted mobility
- **Linear Polymer : Low T_g**

(c) Plasticizers

- Tg is ***inversely proportional*** to plasticization
- **Weaker intermolecular force** between the polymer chains

This is a small molecule which will get in **between the polymer chains**, and space them out from each other



Factors effecting T_g

(d) *Flexibility*

- Tg is **inversely proportional** to flexibility
- Rigid : High T_g
- Flexible Chain: Low T_g

(e) Steric Effect (Substitution):

- Presence of **phenyl group** increase the rigidity, and higher Tg value
- Bulkier Substitution: High T_g
- Smaller Substitution: Low T_g

Factors effecting T_g

(f) **Intermolecular force** is directly proportional to T_g

Presence of '**Hydrogen bonding**' **increase** the T_g

Presence of **electronegative** group increase T_g

(g) **Chain branching**:

- Polymer with **more chain branching** have more free volume, which **reduce T_g**

(h) **Crystallization:**

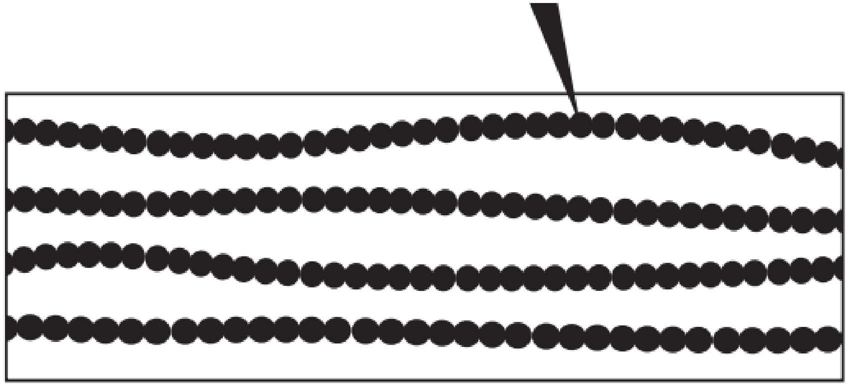
- Crystallization is **directly proportional** to T_g

Effect of Crosslinks on T_g

As a result of the restriction of long-range segmental motion, crosslinking elevates T_g.

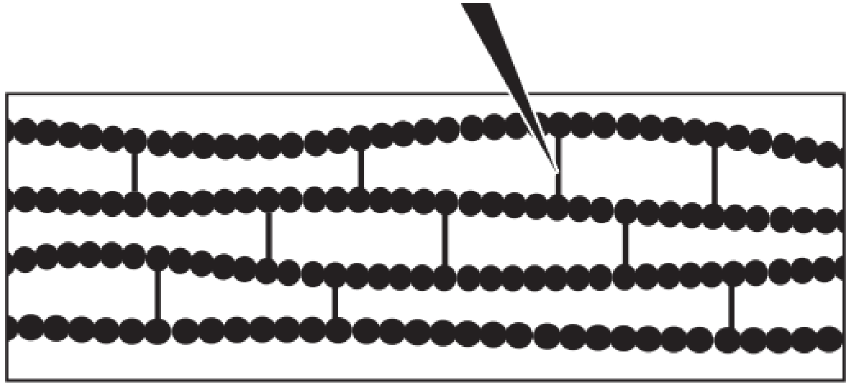
Low T_g

Linear Chain



High T_g

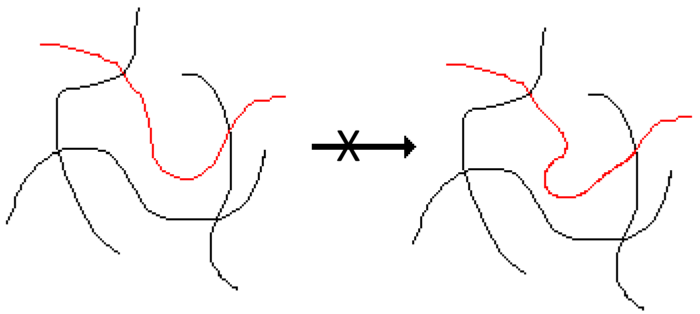
Cross-linked



Molecular Motions Below and Above Tg

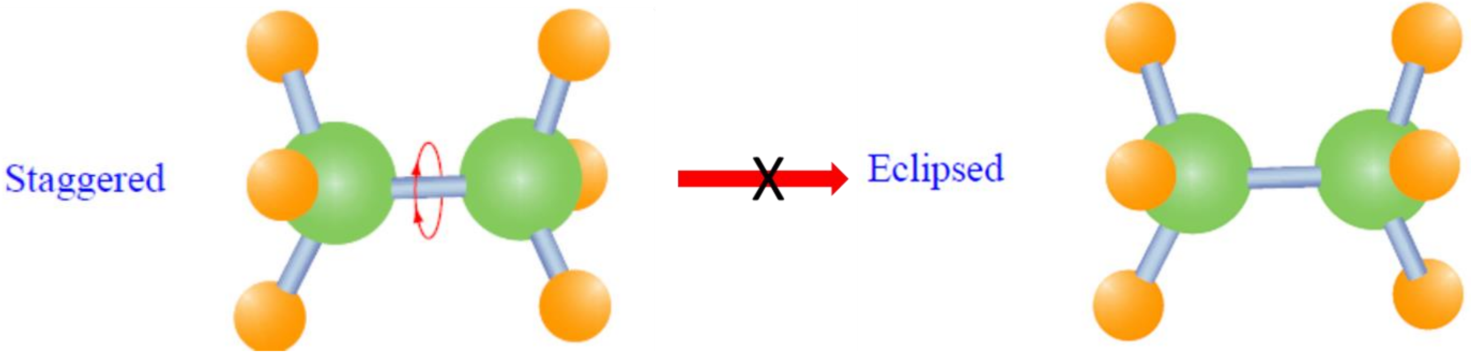
At a low temperature the amorphous regions of a polymer are in the glassy state. In this state the molecules are frozen on place.
Below Tg, wiggling motion or ***segmental motion of polymer chain is not allowed***.

Below Tg



Above Tg

Below T_g, the polymer chains ***do not undergo rotational motions across the bonds***.



*"**Above Tg**, these restricted monitions are thermally activated and the polymer **becomes rubbery**"*

Rotational Motions & T_g

In particular, *any factor that restricts rotational motion within the chain should raise* T_g.

– the more mobile the chain, the lower the value of T_g”

General Trends of T_g

$T_g \propto$ *Strength of intermolecular force* (or appropriately *interchain force*)

$T_g \propto$ *Degree of crystallinity*

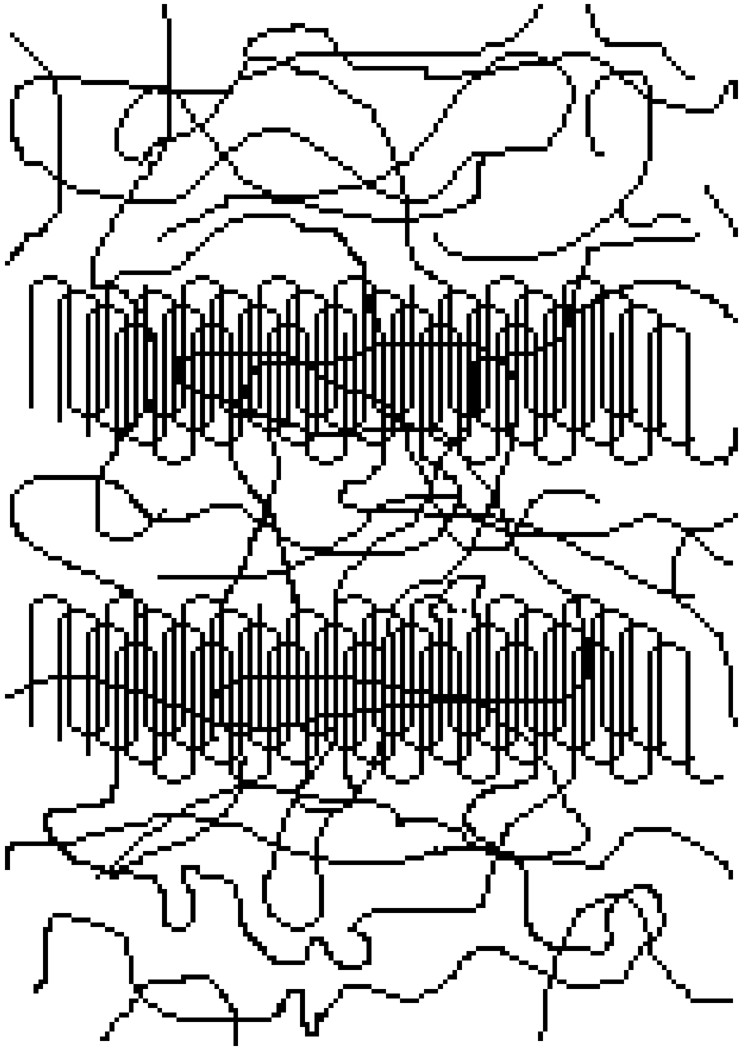
$T_g \propto$ *Degree of cross-linking*

$T_g \propto$ *Rigidity of the polymer backbone and stiffening groups*

$$T_g \propto \frac{1}{\textit{Flexibility of a polymer main chain and branches}}$$

$$T_g \propto \frac{1}{\textit{Extent of rotational motions within the polymer chain}}$$

T_g of Semi-Crystalline Solids



Semi-crystalline solids have both amorphous and crystalline regions.

The glass transition is a property of only the amorphous portion of a semi-crystalline solid.

The crystalline portion remains crystalline during the glass transition.

Semicrystallinity: effect of polymer structure

The degree of crystallinity of the polymer depends on its **structure**.

- Linear polymer like **HDPE** (high-density polyethylene) is more crystalline than (low-density polyethylene) **LDPE**

(i.e., branched polyethylene).

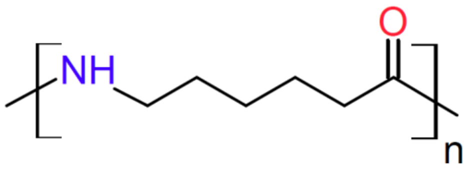
Semicrystallinity: effect of polymer structure

The degree of crystallinity of the polymer depends on its structure.

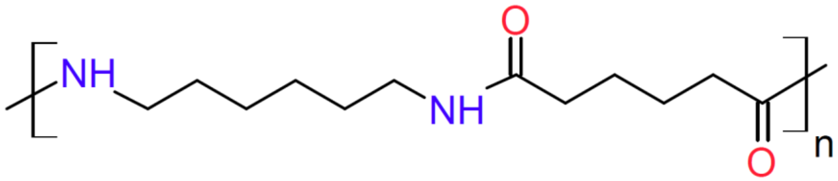
- The polymer chains with **polar groups** can form **hydrogen bonding** with their neighbouring chains and have high crystallinity.

For example, nylon-6,6

Nylon 6 :



Nylon 6,6 :

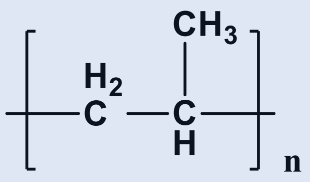


Semicrystallinity: effect of polymer structure

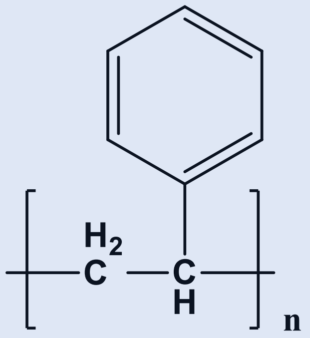
The degree of crystallinity of the polymer depends on its structure.

- Crystallinity of a polymer also depends on the **stereo-regular arrangement**.
- Polymers like isotactic and syndiotactic polystyrene, etc., are highly crystalline.
- On the other hand, atactic polystyrene, in which the substituents are arranged in a random manner, is less crystalline.

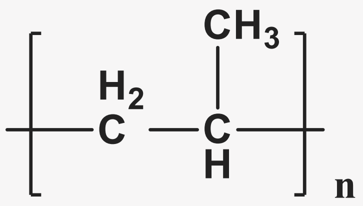
QN



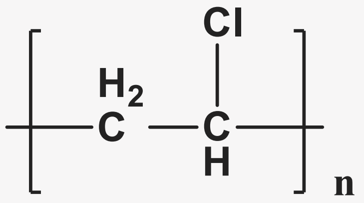
T_g = -20 ^{0}C



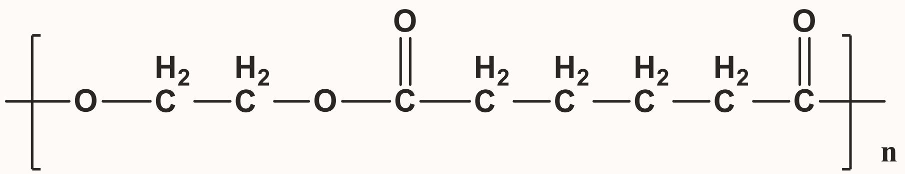
T_g = 81 ^{0}C



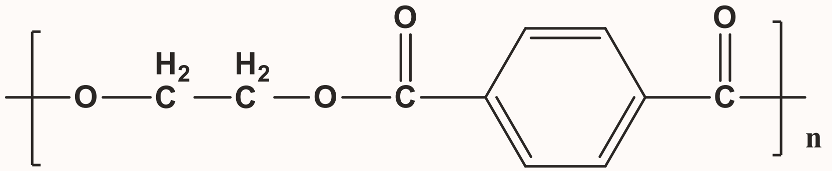
T_g = -20 ^{o}C



T_g = 100 ^{o}C



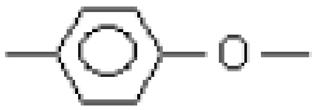
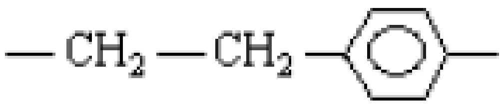
T_g = - 70 ^{0}c

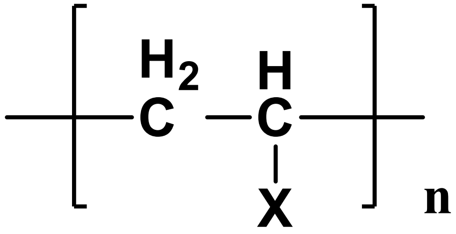


T_g = 69 ^{0}C

Chain Flexibility & Rigidity

Tg is **inversely proportional** to flexibility

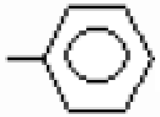
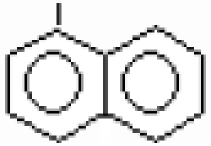
Repeat unit	*name*	T_g	
$—CH_2—CH_2—$	*polyethylene*	180 K	**-93.15°C**
$—CH_2—CH_2—O—$	*poly(oxyethylene)*	206 K	**-67.15°C**
	*poly(phenylene oxide)*	363 K	**89.85°C**
	*poly(p-xylylene)*	353 K	**79.85°C**

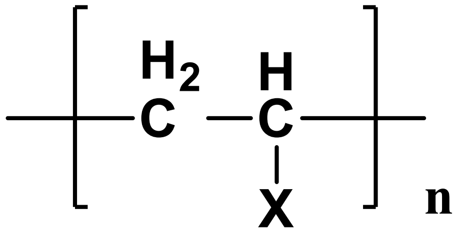


Steric Effects

Bulkier Substitution: High T_g

$$—[CH_2 — CHX]—$$

X =	$—H$	*polyethylene*	T_g =	180 K	**-93.15°C**
	$—CH_3$	*polypropylene*		253 K	**-20.15°C**
		*polystyrene*		373 K	**99.85°C**
		*poly(α-vinylnaphthalene)*		408 K	**134.85°C**



Effect of Intermolecular Forces

Intermolecular force is directly proportional to T_g

X =	$-CH_3$	*polypropylene*	T_g =	253 K	**-20.15°C**
	$-Cl$	*poly(vinyl chloride)*		354 K	**80.85°C**
	$-OH$	*poly(vinyl alcohol)*		358 K	**84.85°C**

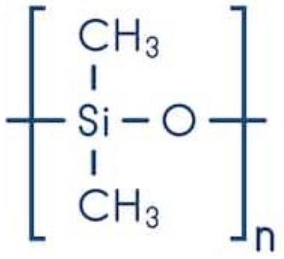
Molecular Structures and T_g

"The glass-transition temperature of amorphous polymers can vary widely with the chemical structure of the polymer chain"

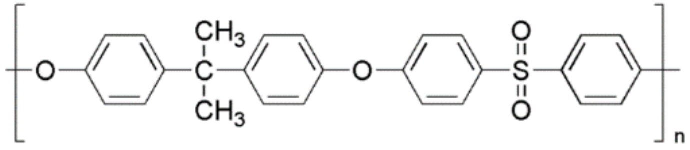
In general, **polymers with flexible backbones and small substituent** groups (e.g., *polyethylene and polydimethylsiloxane*) have low T_g,
while **those with rigid backbones, such as polymers containing main-chain** aromatic groups (e.g.,*polysulfone*), have high T_g.

Table 4-2 Glass-Transition Temperatures of Some Amorphous Polymers

Polymer	T_g (°C)
Polydimethylsiloxane	-123
Poly(vinyl acetate)	28
Polystyrene	100
Poly(methyl methacrylate)	105
Polycarbonate	150
Polysulfone	190
Poly(2,6-dimethyl-1,4-phenylene oxide)	220



Polydimethylsiloxane



Polysulfone

Polymers Classification
-On mechanical Properties

Elastomers, Plastics, and Fibers

Organic Polymers

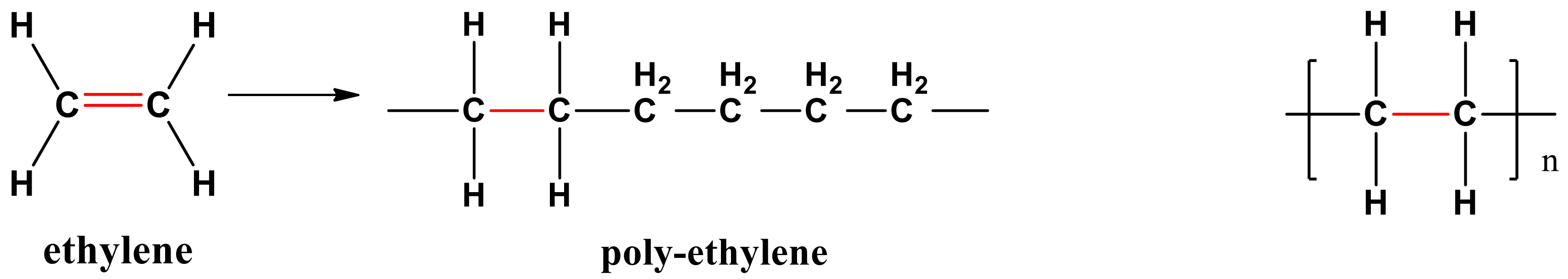
1. Polyethylene (PE)
   (i) LDPE
   (ii) HDPE
2. Polyvinyl chloride (PVC)
3. Polytetrafluoroethylene (PTFE) or Teflon
4. Acrylonitrile-Butadiene-Styrene (ABS)
5. Phenolic resins: Bakelite
6. Nylon-6,6
7. Kevlar

Synthesis

Properties

Uses

(1) Polyethylene



2 types of polyethylene polymer are available :

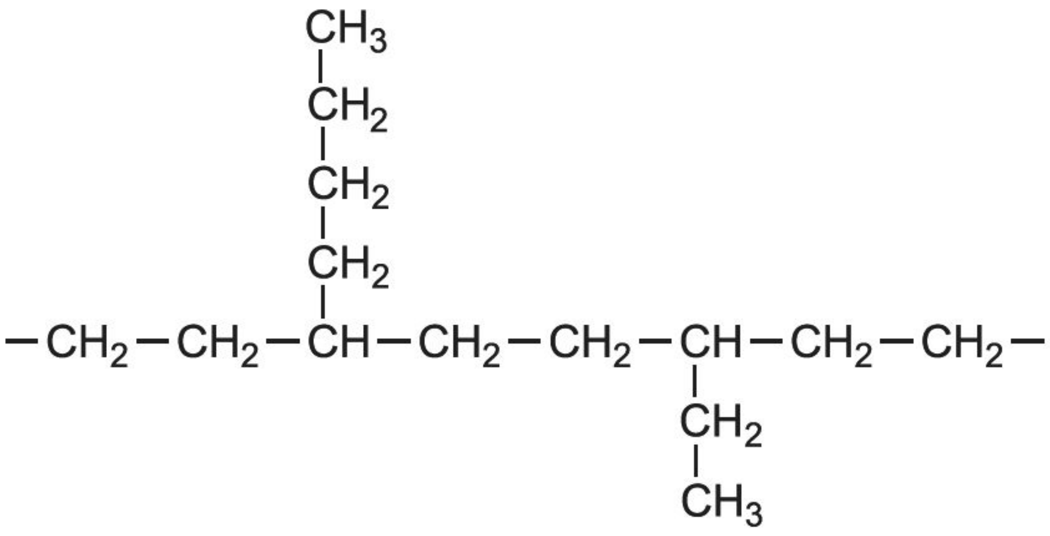
1. Low Density polyethylene (**LDPE**) :

   Produced using high pressure method & **free radical initiator**

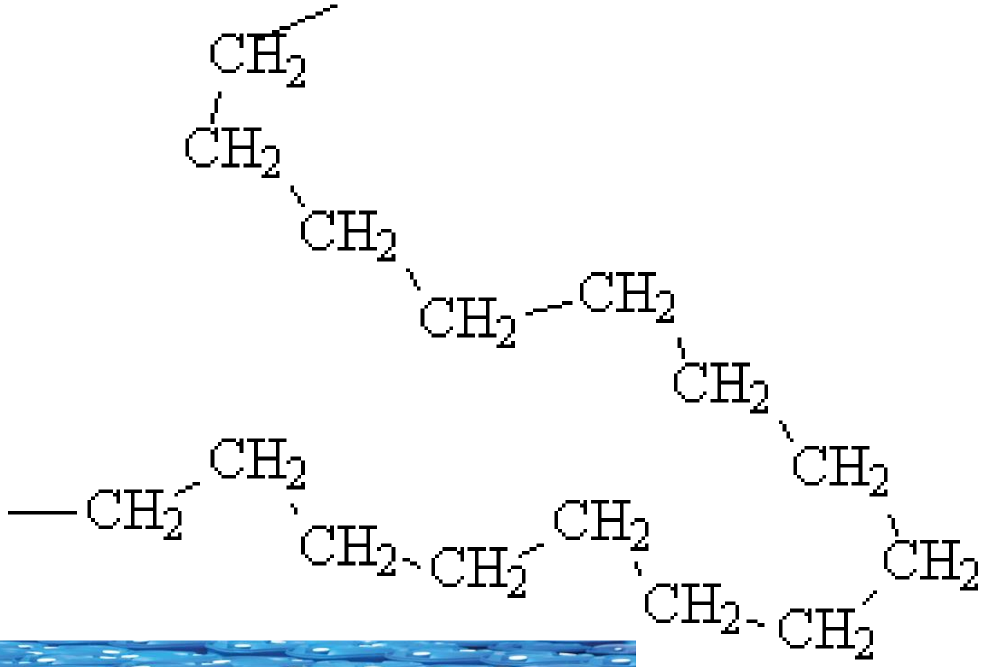
2. High Density polyethylene (**HDPE**) :

   Produced using low pressure method & **ionic catalyst**

Low Density polyethylene (LDPE)



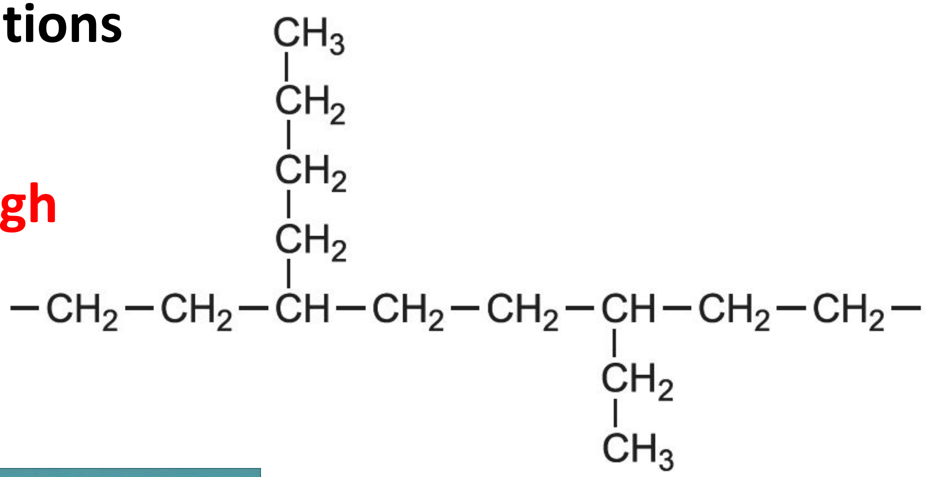
High Density polyethylene (HDPE)



Properties of Polyethylene

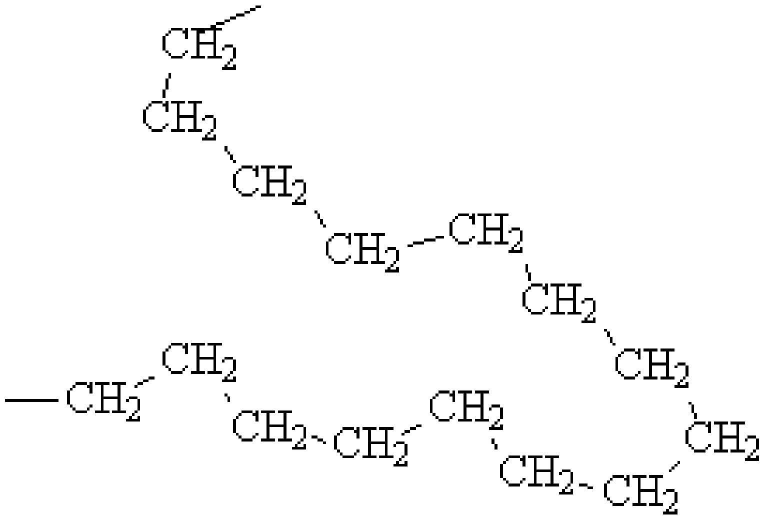
LDPE

- Rigid, waxy, White, translucent non-polar material
- Good chemical resistance against acids, alkalies and salt solutions
- Good insulating property.
- LDPE has branched structure, and hence it is flexible and tough
- Low density



HDPE

- **Linear polymer**
- **Better chemical resistance**
- **Higher softening point**
- **Greater rigidity but low impact strength**
- **Relatively brittle**
- **Polymer chain packing is efficient and density is high with respect to LDPE.**



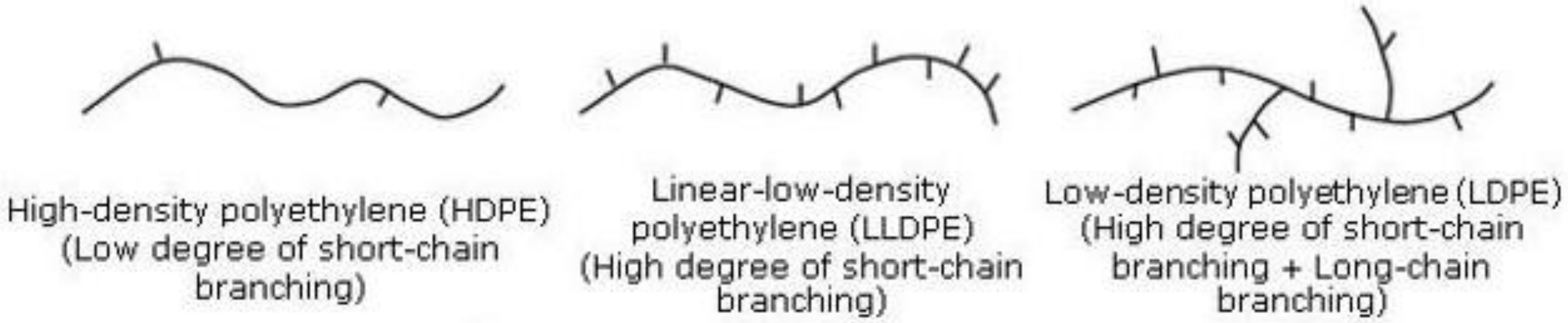
High Density Polyethylene (HDPE)

LDPE	HDPE
Formed under *high pressure* with the use of *radical initiator*.	Formed under *low pressure* with the help of an *ionic catalyst (Lew acid)*
The use of high pressure and free radical initiator makes reaction faster. This leads to *high level of branching.*	The use of low pressure and the ionic catalyst moderates the velocity of the reaction. This leads to *negligible extent of branching.*
High level of branching results in lower polymer chain packing. Consequently polymer chain *packing density is low.*	Branching is negligible. Hence, polymer chain *packing density* is high.
Degree of crystallinity is low. Hence, *Tg is low.*	Degree of crystallinity is high. Hence, *Tg is high.*
Transparent	Opaque
Flexible and tough	Hard and brittle

HDPE has high Tg, whereas LDPE has low Tg. Why?

QN If the three type of polyethylene shown below are processed under similar conditions, the lowest melting (Tm) temperature is possessed by

a) HDPE
b) LLDPE
c) LDPE
d) They all have the same Tm.



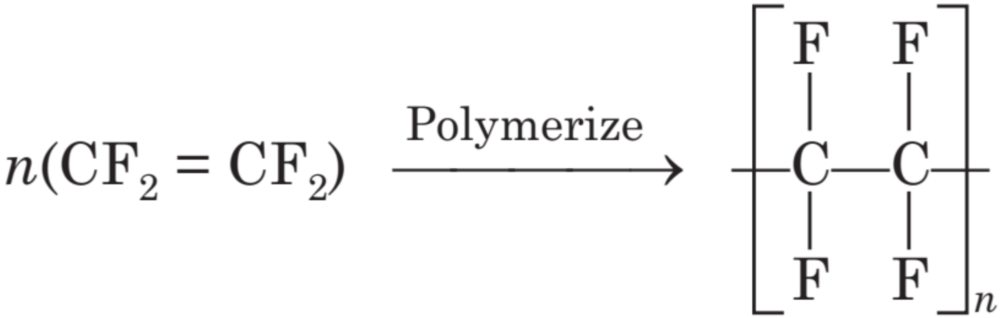
Note: Here the thermal transition referred as Tm,
because *all type of polyethylene will have significant level of crystalline order.*

(2) Polyvinyl chloride (PVC)

- **Synthesis**: $$nCH_2=CHCl \xrightarrow{\text{Polymerize}} \underset{\text{PVC}}{-\!\!(CH_2CHCl)_n\!-}$$
- **Properties**:
  - Colourless, odourless, non-flammable and chemically inert powder
  - High rigidity, brittle, and chemical resistance
  - It contain 53-55 % chlorine, and softens at around 80^0C
  - Resistant to water, light, O_2, inorganic acid and alkalies, oil, petrol
  - Soluble in hot chlorinated hydrocarbons
- **Uses**
  - Most widely used Plastics.
  - Cables, water hoses, toys, rain coats, rexin, pipes, floor covering, refrigerator components, tyres, cycles, motor cycle mudguards, etc.

(3) Polytetrafluoroethylene (PTFE) or Teflon

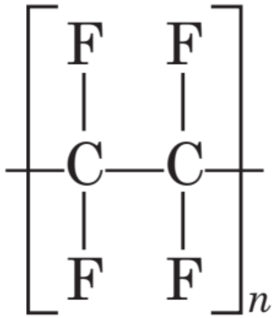
- **Synthesis:**



- **Properties:**
  - Due to the presence of **highly electronegative** fluoride, **strong interchain forces** are present, which give the material extraordinary property like **extreme toughness**, **high softening point** (350^0C), **high chemical resistance, low coefficient of fabrication and waxy touch.**
  - **Good mechanical and electrical properties.**

Polytetrafluoroethylene (PTFE) or Teflon

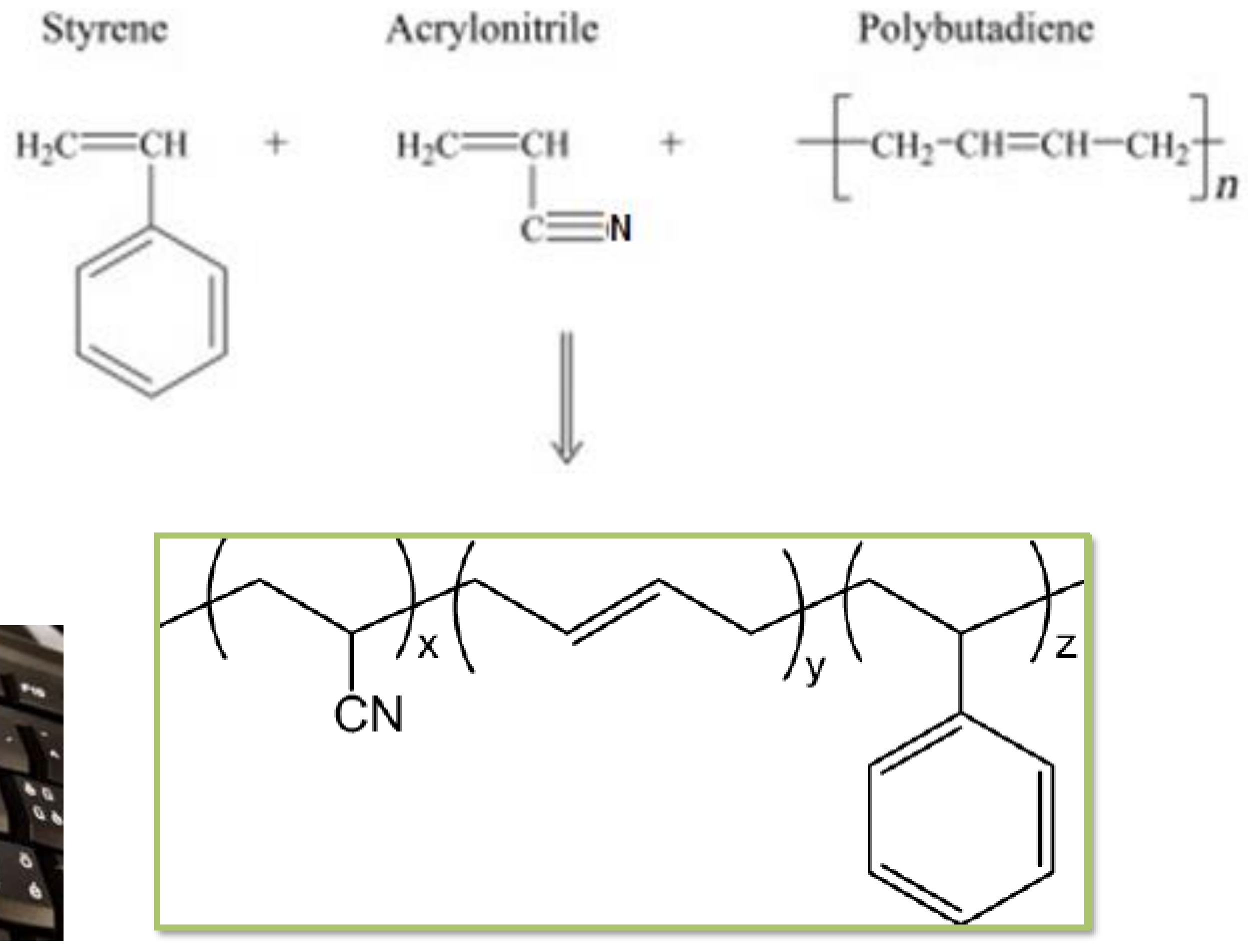
- **Uses:**
  - **Insulating motor, transformers, cables, wires, etc**
  - **Non-stick cookware**
  - **Making gaskets, pump parts, tank lining**
  - **Non-lubricating bearings**



(4) Acrylonitrile-Butadiene-Styrene (ABS)

Synthesis:

It is a polymer made by dissolving polybutadiene in liquid acrylonitrile and styrene monomers and then polymerizing the monomers by the introduction of free-radical initiators.

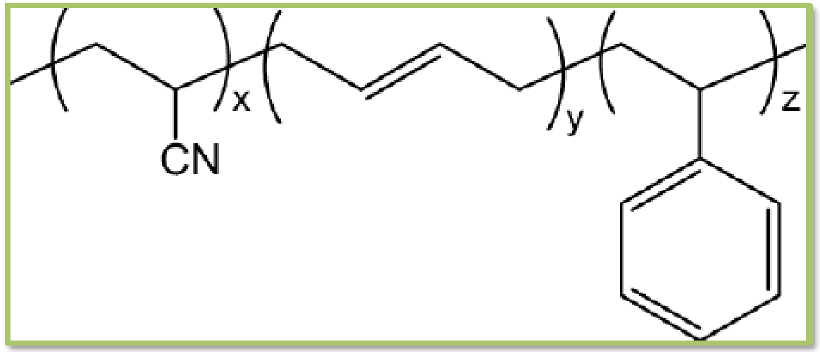


Copolymer

Acrylonitrile-Butadiene-Styrene (ABS)

Properties:

- ABS is a tough, **heat-resistant thermoplastic**.
- **styrene provides** ease of processing, glossiness, and rigidity.
- **Acrylonitrile** adds chemical resistance and hardness,
- **butadiene** provides impact resistance.



Uses:

- ABS is the standard material of choice for the ***strong plastic cases of computers, TV, and other household electronic goods and sometimes is used as an alternative to PVC for pipes.***

Musical instruments such as recorders, plastic oboes and clarinets, piano movements, and keyboard keycaps are commonly made out of ABS

(5) Phenolic resins: Bakelite

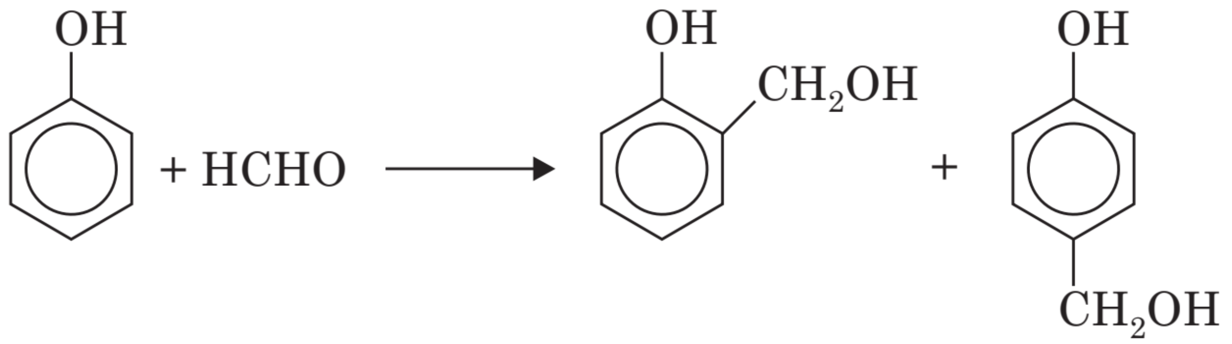
- Bakelite was the **first plastic** made from synthetic components.
- It is a thermosetting phenol formaldehyde resin, formed from a condensation reaction of phenol with formaldehyde.

Phenolic resins: Bakelite

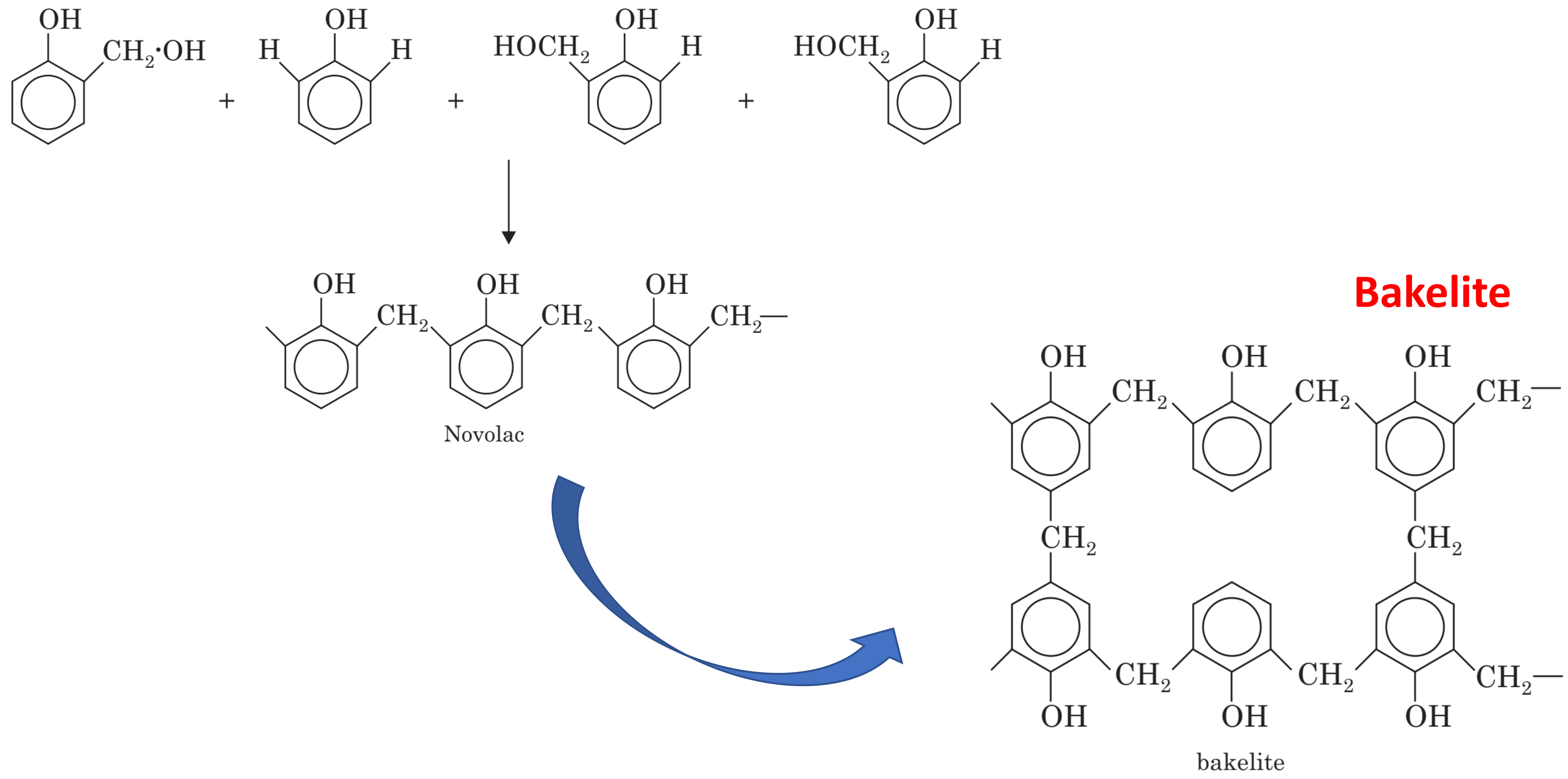
Synthesis:

- Prepared by **condensation** of phenol with formaldehyde in presence of acid-alkaline catalyst in an aqueous solution.

- In the presence of acid catalyst, the 1st step leads to the formation of **ortho**- and **para**-hydroxy methyl phenol, which form **linear polymer**.



Synthesis of Bakelite



Bakelite

Bakelite Cont.



- **Properties**:
  - The phenolic resins are **rigid**, **hard**, **water resistant**.
  - Resistant to non-oxidising acids, organic solvents, but susceptible to alkalies
  - Solubilities and Melting point of the resin gradually change with rise if molecular weight.
  - Resins are **electrically insulator**

- **Uses**:
  - Widely used as metal substitute where high tensile strength is not necessary.
  - It is used for making insulator part like **switches, plug, heater handles**
  - It can be moulded to cabinets for TVs and radio and telephone parts
  - It is used as Adhesive in **paints and varnishes**
  - As cation exchanger resin for **water softening**
  - In paper industry as propeller shafts


6. Fibers: Nylon-6,6

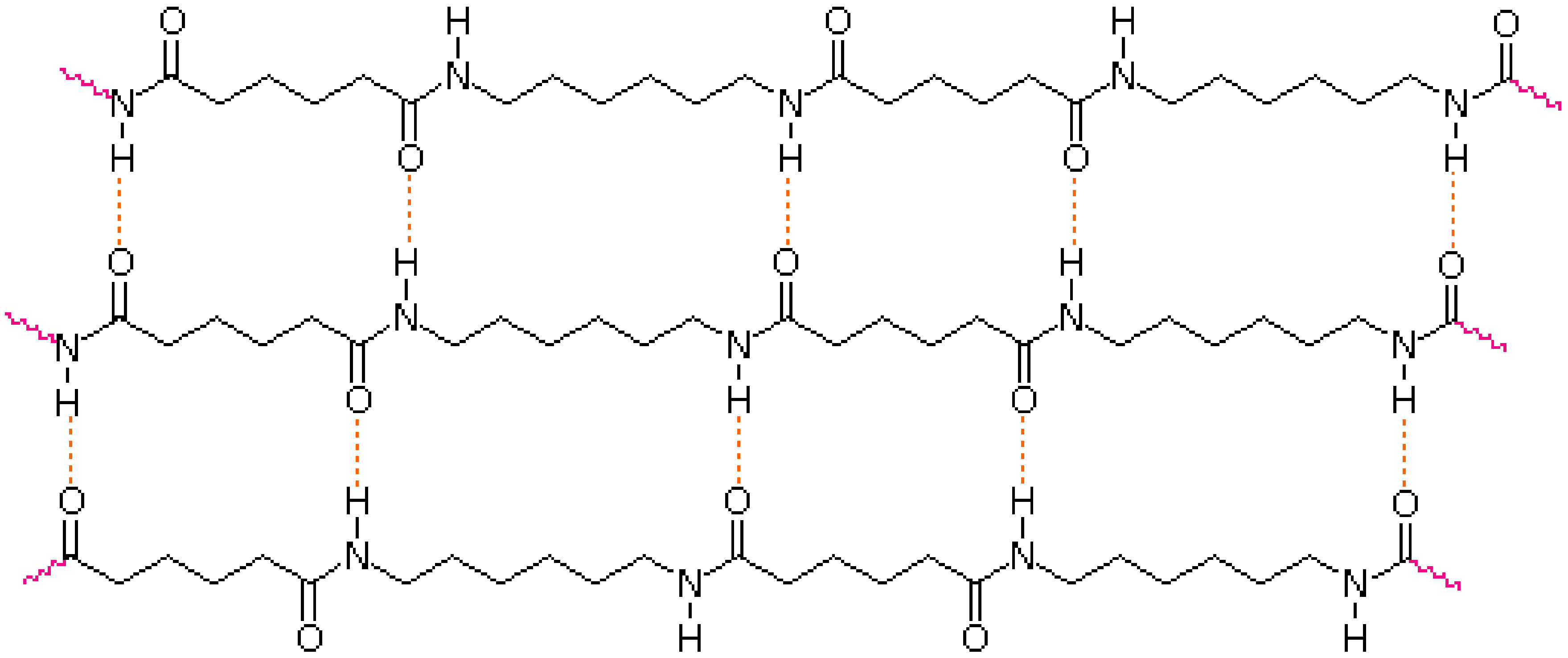
- **Nylon-6,6** is a type of polyamide or nylon.
- It, and nylon 6, are the two most common for textile and plastic industries.
- Nylon-6,6 is made of two monomers each containing 6 carbon atoms, hexamethylenediamine and adipic acid, which give nylon-6,6 its name.

$$\underbrace{n\ HO-\overset{\overset{\displaystyle O}{\|}}{C}-(CH_2)_4-\overset{\overset{\displaystyle O}{\|}}{C}-OH}_{\text{adipic acid}} + \underbrace{n\ H_2N-(CH_2)_6-NH_2}_{\text{hexamethylenediamine}} \longrightarrow$$

$$\underbrace{\left[\overset{\overset{\displaystyle O}{\|}}{C}-(CH_2)_4-\overset{\overset{\displaystyle O}{\|}}{C}-NH-(CH_2)_6-NH\right]_n}_{\text{nylon-6,6}} + 2n\ H_2O$$

H-Bonding in Nylon-6,6

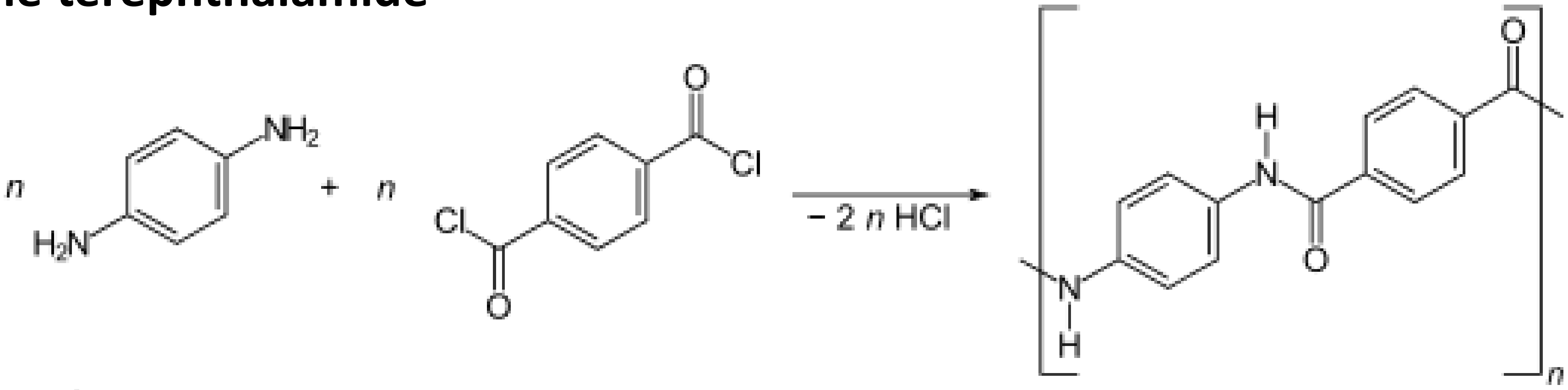
Interchain Hydrogen Bonding Enhances Crystallinity

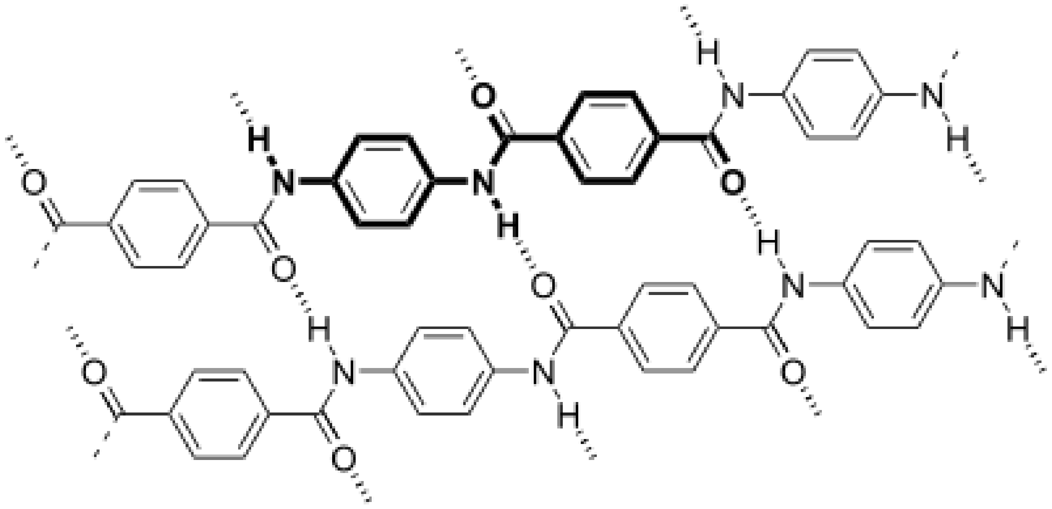


7. Kevlar

- **Kevlar** is a heat-resistant and strong synthetic fiber.
- It is a **Poly-paraphenylene terephthalamide**

Synthesis:



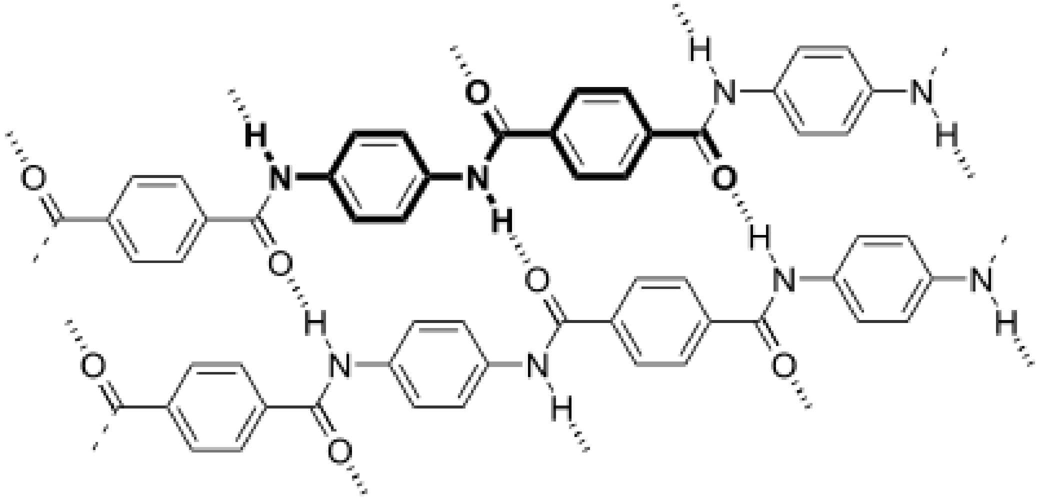


Kevlar

Properties:

- When **Kevlar** is spun, the resulting fiber has **high** ***tensile strength*** of about 3,620 MPa, and a relative **density** of 1.44.

- The polymer owes its **high strength** to the many ***inter-chain hydrogen bonds***. *The fiber is used to make* ***bulletproof vests***



QN

Match the following!

Linear Chain Polymer	Polyester
Branched Polymer	Bakelite
Heterochain polymer	Low density polyethylene (LDPE)
Thermoset	Polystyrene
Homochain polymer	High density polyethylene (HDPE)
Polyethylene terephthalate	Nylon 6,6

Match the following!

Linear Chain Polymer	Polyester
Branched Polymer	Bakelite
Heterochain polymer	Low density polyethylene (LDPE)
Thermoset	Polystyrene
Homochain polymer	High density polyethylene (HDPE)
Polyethylene terephthalate	Nylon 6,6

Elastomers

Elastomers: The polymers which undergo high degree of elongation when pulled apart, and return to their original length on release are called elastomers.

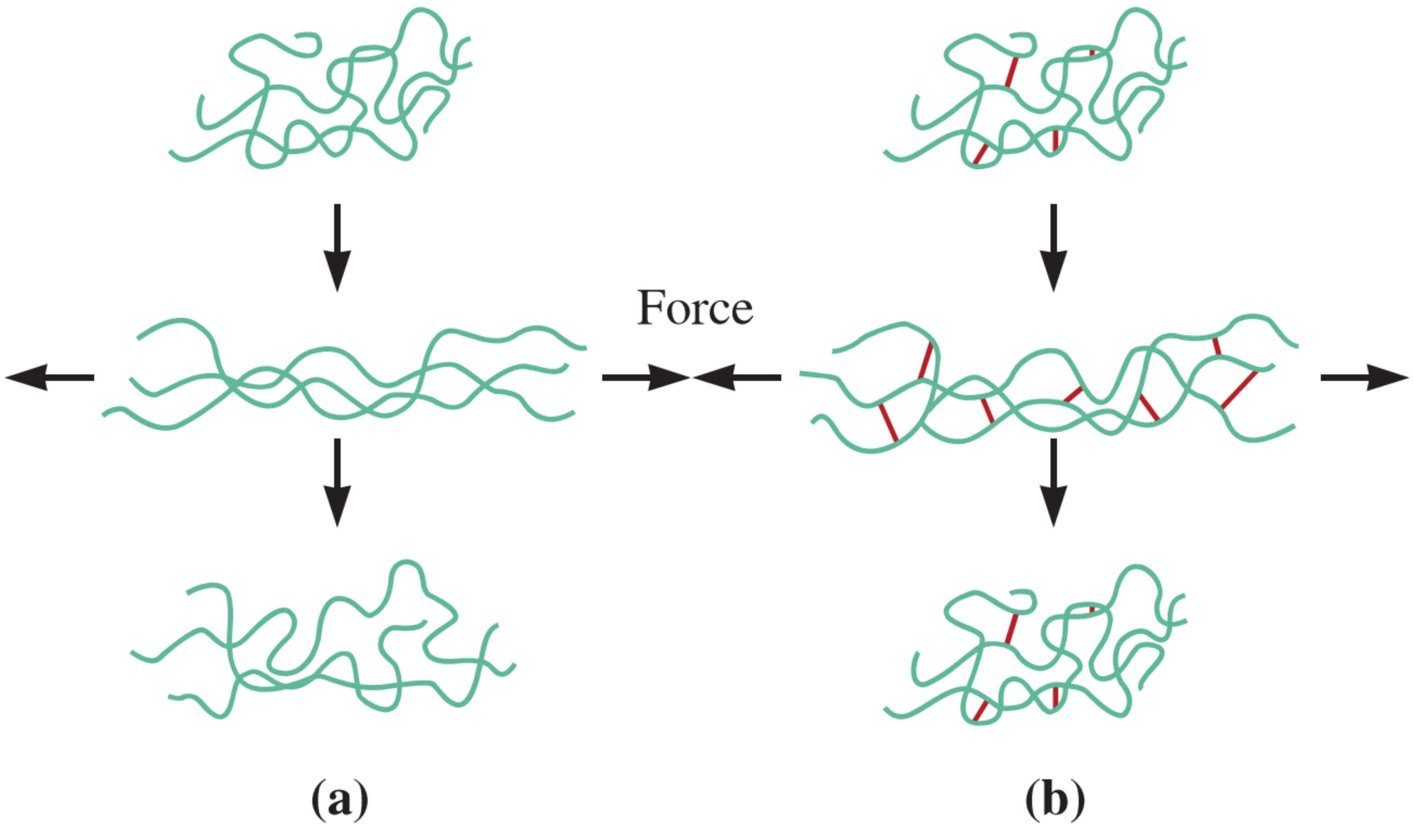
These are mainly coiled and long-chain polymers with **no intermolecular forces** except weak van der Waals forces (dispersion forces).

For example: **natural rubber**, Buna-S, butyl rubber, silicone rubber, etc.

Polymers must have little internal hindrance to the random motion of their monomer subunits (in other words, they must not be glassy), and they must not spontaneously crystallize (at least at normal temperatures).

On release from being extended, they must be able to return spontaneously to a disordered state by random motions of their repeating units as a result of rotations around the carbon-carbon bond.

Linear *vs* Cross linked elastomers



"Small degree of cross-links in elastomers acts as a memory to return to its original shape and size upon release of stress"

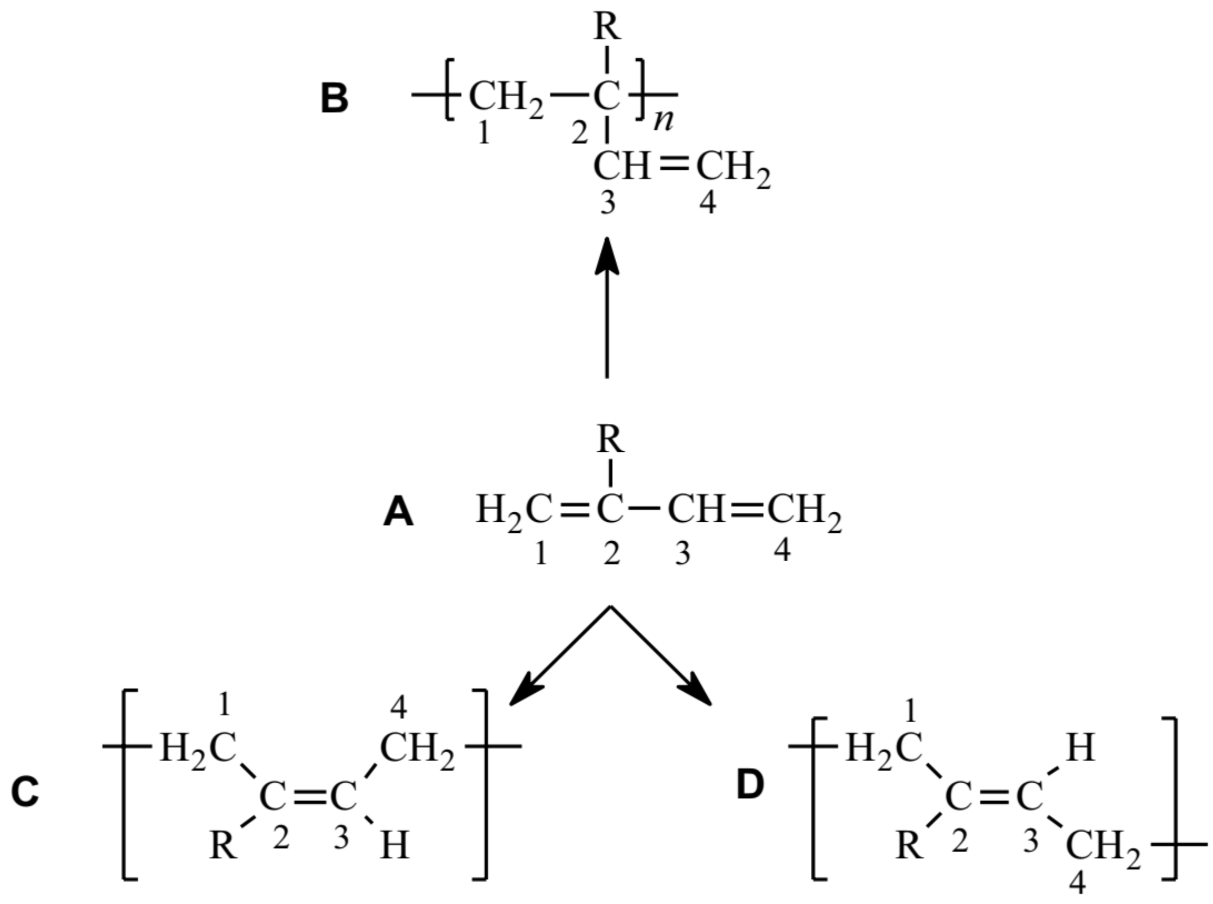
Figure 16-21 (a) When the elastomer contains no cross-links, the application of a force causes both elastic and plastic deformation; after the load is removed, the elastomer is permanently deformed. (b) When cross-linking occurs, the elastomer still may undergo large elastic deformation; however, when the load is removed, the elastomer returns to its original shape.

Elastomers [Rubbers]

A number of natural and synthetic polymers called elastomers display a large amount ($>200\%$) of elastic deformation when a force is applied. Rubber bands, automobile tires, O-rings, hoses, and insulation for electrical wires are common uses for these materials. Crude natural rubber, which is an elastomer, can erase pencil marks; hence, elastomers got the name rubber.

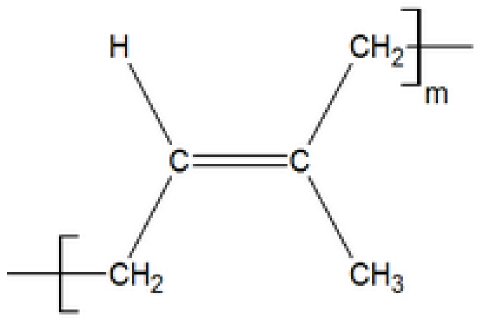
- Elastomers can be broadly classified as belonging to one of three groups:
  - Diene elastomers,
  - Non-diene elastomers, and
  - Thermoplastic elastomers.

Diene Elastomers

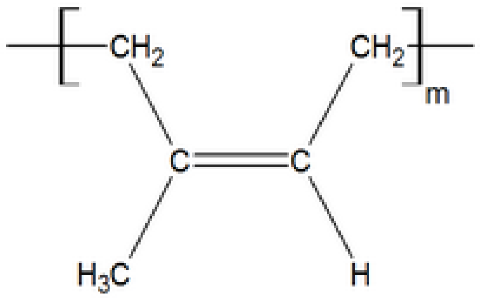


Substituent, R	Monomer	Elastomer
H	1,3-Butadiene	Polybutadiene
Cl	2-Chloro-1,3-butadiene	Polychloroprene
CH_3	2-Methyl-1,3-butadiene	Polyisoprene

Diene Elastomers



Trans isoprene



Cis isoprene

- Trans isoprene structure can
leads to relatively straight chain structure.

- can have crystalline structure (compared
to cis-form)

- less elastomer property

- cis isoprene structure leads to amorphos form

- this can show little elastomer property.

Non-diene Elastomers

- A number of important elastomers do not have the unsaturated chain structure characteristic of the diene elastomer.

- These nondiene elastomers include
  polyisobutylene (butyl rubber),
  polysiloxane (silicone rubber),
  fluoroelastomers such as Viton,
  polyurethane elastomers such as Spandex, and
  elastomers derived from ethylene and propylene (EP and EPDM elastomers).

Plastics

Plastics: These are polymers, which can be moulded into desired shapes by the application of heat and pressure.

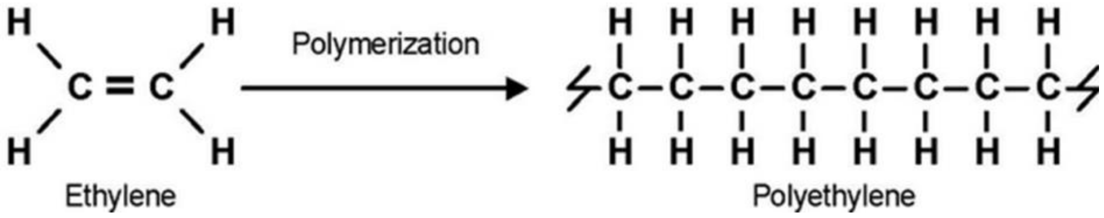
For example: **PE**, **Plexiglass**, **PVC**, **PC**, **Teflon**, etc.

Thermoplastic polymers on application of heat and pressure, initially become soft, flexible and undergo deformation.

On further heating above their melting point (Tm) they melt and flow. Such a property is called **plastic deformation**.

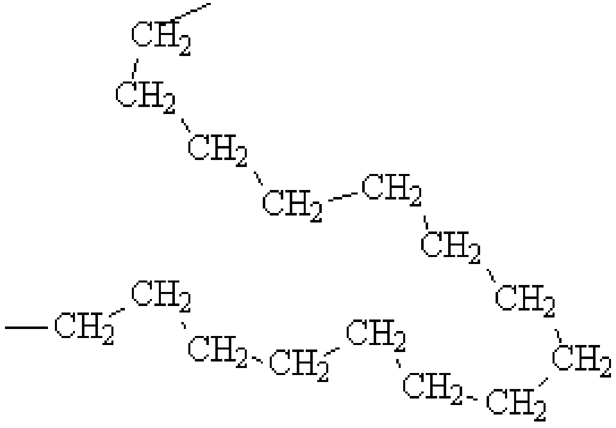
Thermoplastics exhibit plastic deformation because they are **linear**, closely packed and the individual chains are held by secondary forces such as van der Waals force (dispersion), and dipole– dipole interactions.

High density and low density polyethylene



HDPE (High density polyethylene):

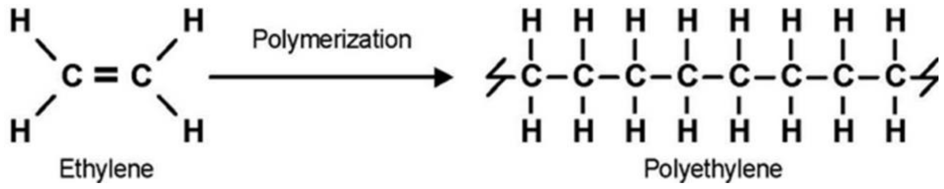
- The polymer chains in HDPE are more linear.
- They pack closer together, resulting in greater intermolecular forces and a more "crystalline" structure.
- HDPE has greater tensile strength than LDPE.
- Density **0.944 – 0.965 g/cm^3**



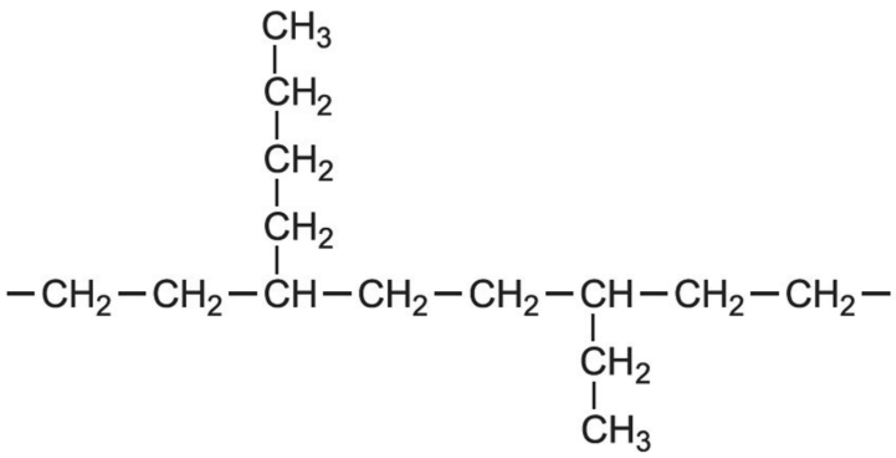
High Density Polyethylene (HDPE)

High density and low density polyethylene

LDPE (Low density polyethylene):



- The polymer chains of LDPE are highly branched
- This branching prevents the chains from stacking neatly beside each other, reducing the intermolecular forces of attraction.
- This results in a plastic that is softer and more flexible, but which also has lower tensile strength.
- Density **0.917 – 0.930 g/cm^3**

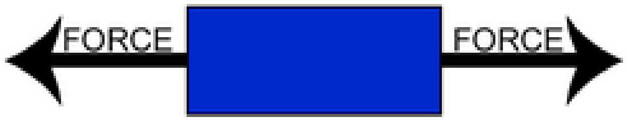


High density and low density polyethylene

Tensile strength

Tensile strength is the ability of a material to withstand a pulling (tensile) force.

It is customarily measured in units of force per cross-sectional area.



The ability to resist breaking under tensile stress is one of the most important and widely measured properties of materials used in structural applications.

Tensile strength is important in the use of brittle materials more than ductile materials.

Polyvinyl chloride (PVC)

- Synthesis:

$$n\mathrm{CH_2=CHCl} \xrightarrow{\text{Polymerize}} \underset{\text{PVC}}{-\!\!\left(\mathrm{CH_2CHCl}\right)\!\!-_n}$$

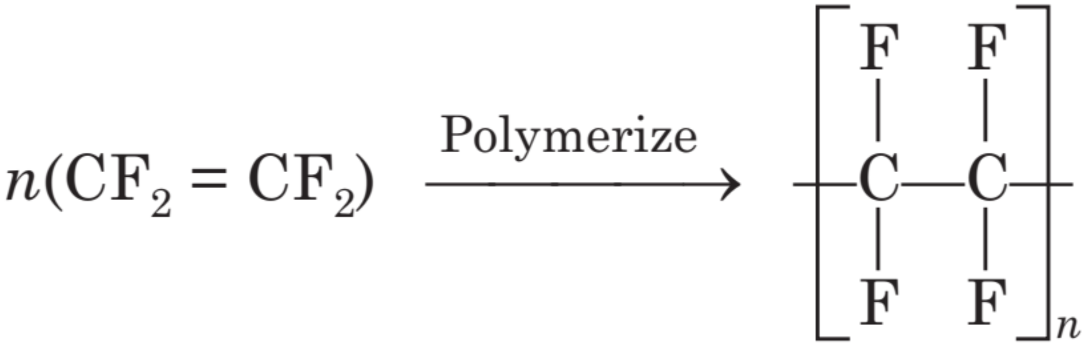
Properties: PVC is a colorless, odorless, non-flammable, chemically-inert powder. It contains 53-55% Cl_2 and softens at around 80°C.

It is resistant to water, light, O_2, inorganic acids and alkalies, oil, petrol etc., but soluble in hot chlorinated hydrocarbons.

Uses. It is the most widely used plastic. It has high rigidity and chemical resistance but brittle, so, its use is mainly in making cables, water hoses, toys, rain coats, rexin, pipes of petroleum industry, floor covering, refrigerator components, tyres, cycles and motor cycle mudguards etc.

Polytetrafluoroethene (PTFE)

- Synthesis:

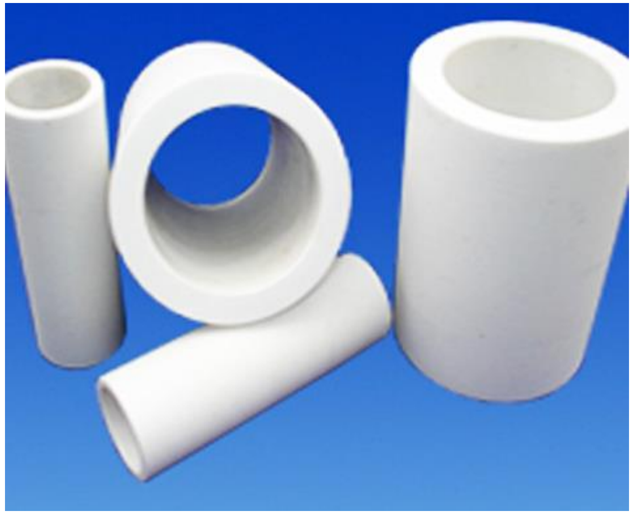


Properties. Due to the presence of highly electronegative fluorine in the regular polymer structure of TEFLON strong interchain forces are present which give the material extraordinary

properties like extreme toughness, high softening point (350°C), high chemical resistance, low coefficient of friction and waxy touch, good mechanical and electrical properties. Due to all these qualities, the polymeric material can be machined. It softens at about 350°C, hence at this high temperature it can be moulded applying high pressure.

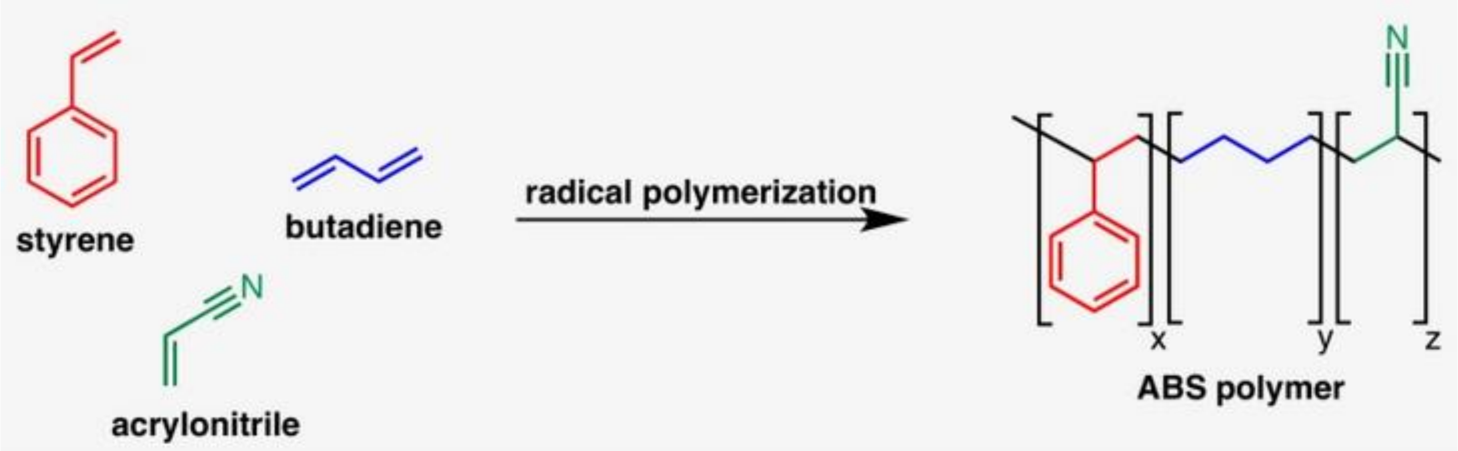
Properties and Uses of PTFE

Uses. It can be used for insulating motor, transformers, cables, wires etc. Non-stick cookware coatings are made of TEFLON. It can also be used for making gaskets, pump parts, tank linings, pipes and tubes for chemical industry, non-lubricating bearings and to make non-reactive coating.



Acrylonitrile-Butadiene-Styrene (ABS)

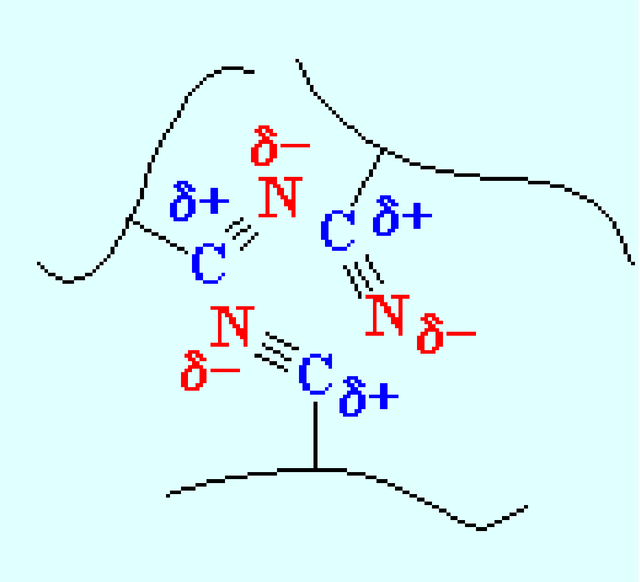
It is Graft polymer made by dissolving polybutadiene in liquid acrylonitrile and styrene monomers and then polymerizing the monomers by the introduction of free-radical initiators.
It is a giant molecule predominantly made up of chains of polybutadiene backbone with styrene-acrylonitrile copolymer (SAN) branches.



ABS is a tough, **heat-resistant thermoplastic**.
Polystyrene provides ease of processing, glossiness, and rigidity.
Acrylonitrile adds chemical resistance and hardness,
and **butadiene** provides impact resistance.

ABS is the standard material of choice for the ***strong plastic cases of computers, televisions, and other household electronic goods and sometimes is used as an alternative to PVC for pipes.***

Acrylonitrile-Butadiene-Styrene (ABS)



ABS is a stronger plastic than polystyrene because of the nitrile groups of its acrylonitrile units.

The nitrile groups are very polar, so they are attracted to each other. This allows opposite charges on the nitrile groups to stabilize each other like you see in the picture on the left.

This strong attraction holds ABS chains together tightly, making the material stronger. Also the rubbery polybutadiene makes ABS tougher than polystyrene.

Fibers

These are long, thin and thread-like polymers, whose length is at least 100 times their diameter.
They do not undergo stretching and deformation like elastomers, and are linked to each other by hydrogen bonding.

In fibers, the chain mobility is reduced by very close packing of the polymer chain backbone without cross-linking.

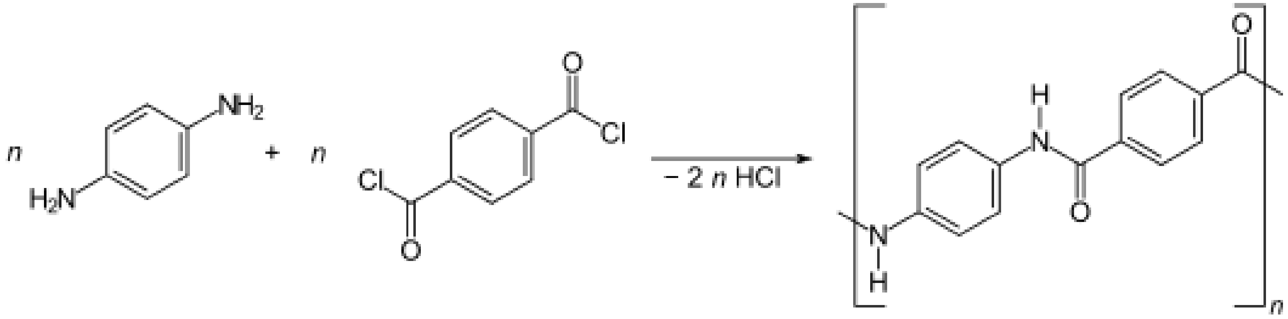
Polar groups and aromatic and cyclic rings in the backbone chain are known to impart a high strength to the polymer fiber.

For example, natural fibers such as jute, wood, silk, etc. and synthetic fibers such as nylon-6,6 and terylene.

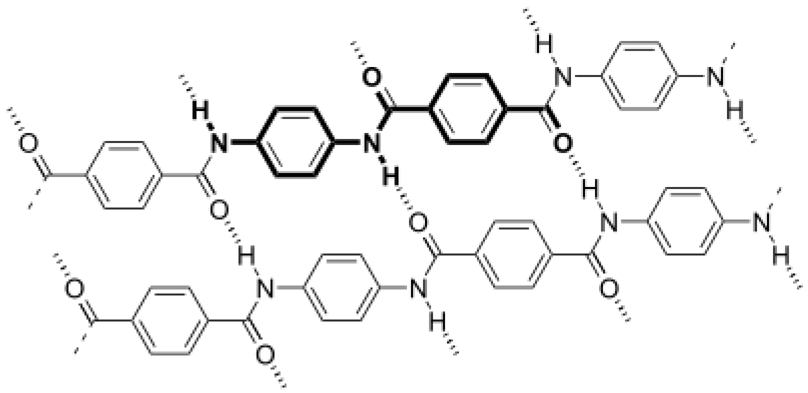
Fibers - Kevlar Fiber

Kevlar is a heat-resistant and strong synthetic fiber.

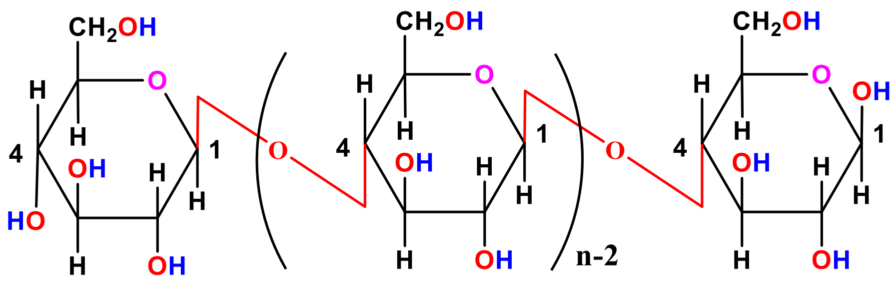
Synthesis

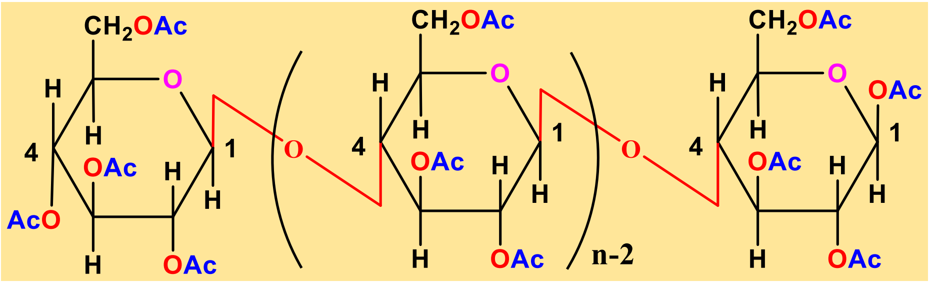


They do not undergo stretching and deformation like elastomers, and are linked to each other by hydrogen bonding.
The fiber is used to make bulletproof vests.



Fibers – Cellulose Fiber





Ac denotes acetyl, $COCH_3$

Cellulose

Cellulose Acetate

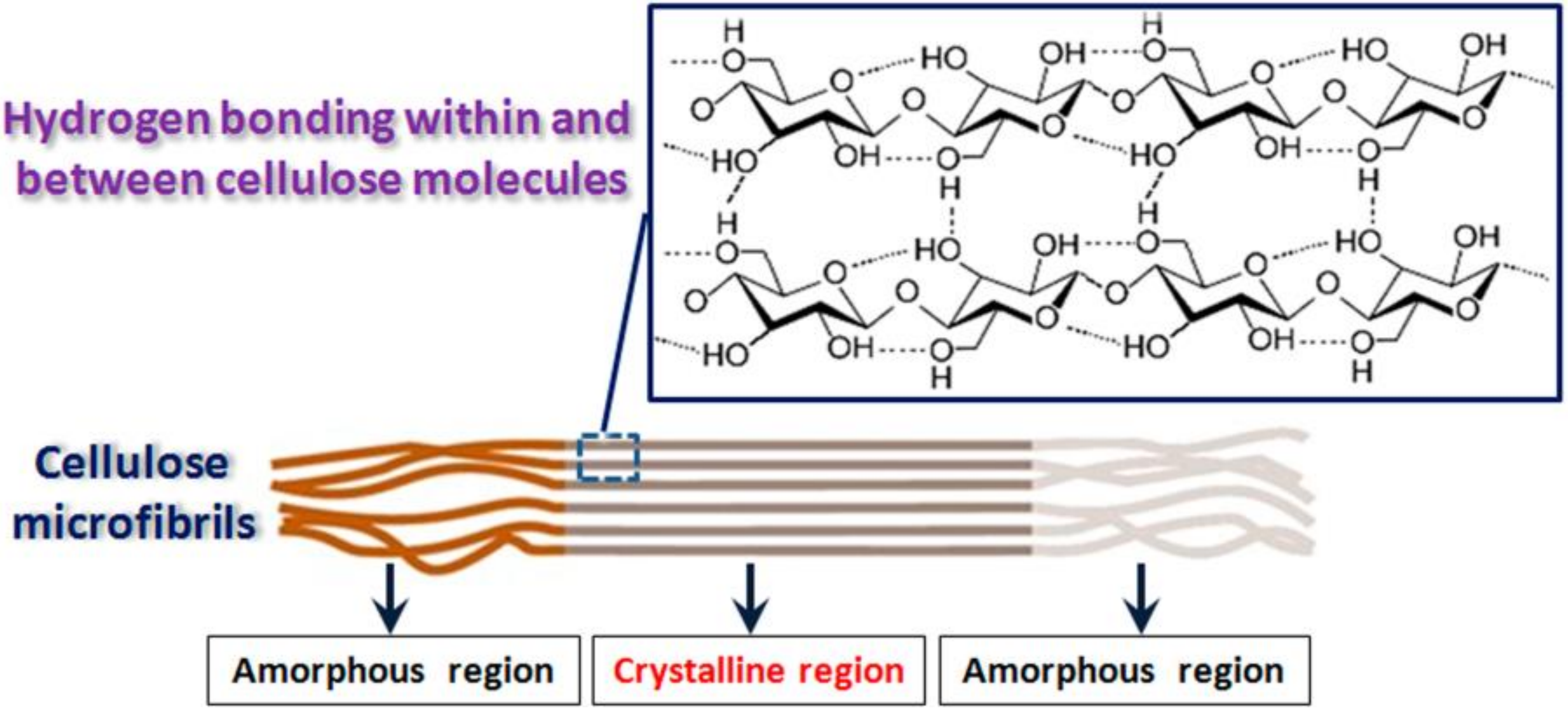
Hydrogen bonding is more

Rigidity is more, and has more fibrous character

No hydrogen bonding

No fibrous character

The β *(1 → 4) -glycosidic linkages* enables *the cellulose polymer to form* ***intermolecular hydrogen bonding*** *with the* ***adjacent cellulose polymer units****.* Large numbers of cellulose polymers blend together via intermolecular hydrogen bonding to form ***fibrous structures***. Despite cellulose is usually classified as a crystalline polymer, it has alternating crystalline and amorphous regions. The high degree of intermolecular hydrogen bonding and fibrous structures makes cellulose an insoluble molecule in water and common organic solvents.



Classification of Polymers

Chemical Structure

Polymeric structure

Conductivity

- Conducting
- Non-conducting

Thermal Behaviour

Molecular Force

Methods of Synthesis

Origin

Conductivity

Conducting polymer

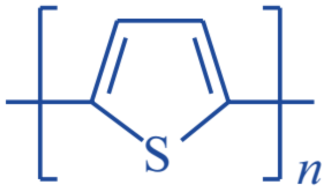
- ***Alternate double bond***
- ***Trans compound is more conductive than cis***

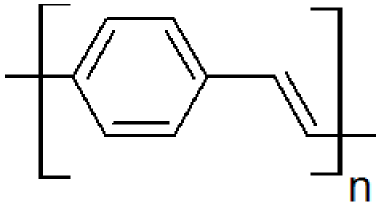
Non-conducting polymer

Absence of Alternate double bond

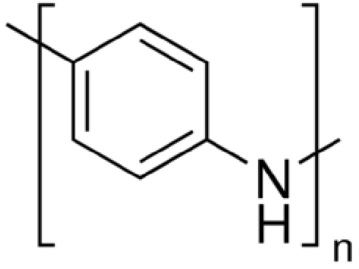
✓ organic polymers conduct electricity *due to the extended conjugation of π-electron clouds*

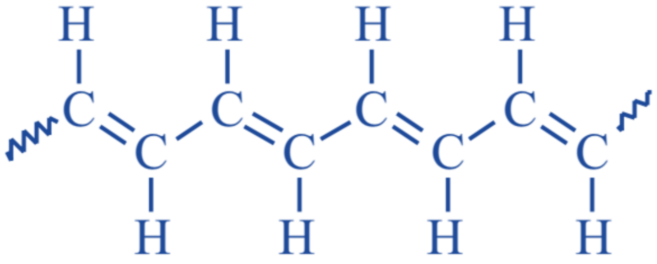
✓ ***Alternate double bond***





Polyphenylene Vinylene





trans-polyacetylene

Applications:

(a) polymeric electrodes for lightweight batteries

(b) variable-transmission windows

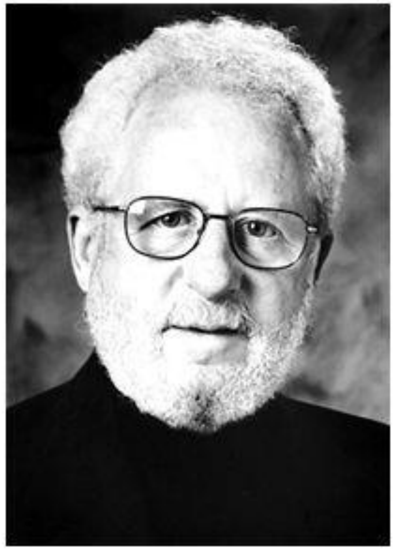
(c) Electrochromic displays

(d) Sensors

(d) Nonlinear optical (NLO) materials

Can plastics conduct?

- Alan J. Heeger, Alan G. MacDiarmid and Hideki Shirakawa made a breakthrough in 1977
- Nobel Prize in Chemistry in 2000 for the discovery and development of electronically conductive polymers



Conductive Polymer

- Polymers obtained by the usual methods of polymerization are mainly insulators.
- A number of polymers are electrically conductive or can be made to be conductive by doping with an electron donor or acceptor **(n-type or p-type dopants).**
- **Applications:**

(a) polymeric electrodes for lightweight batteries,

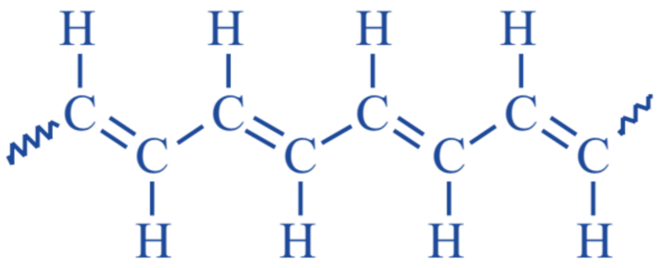
(b) variable-transmission windows,

(c) electrochromic displays,

(d) sensors

(e) nonlinear optical (NLO) materials.

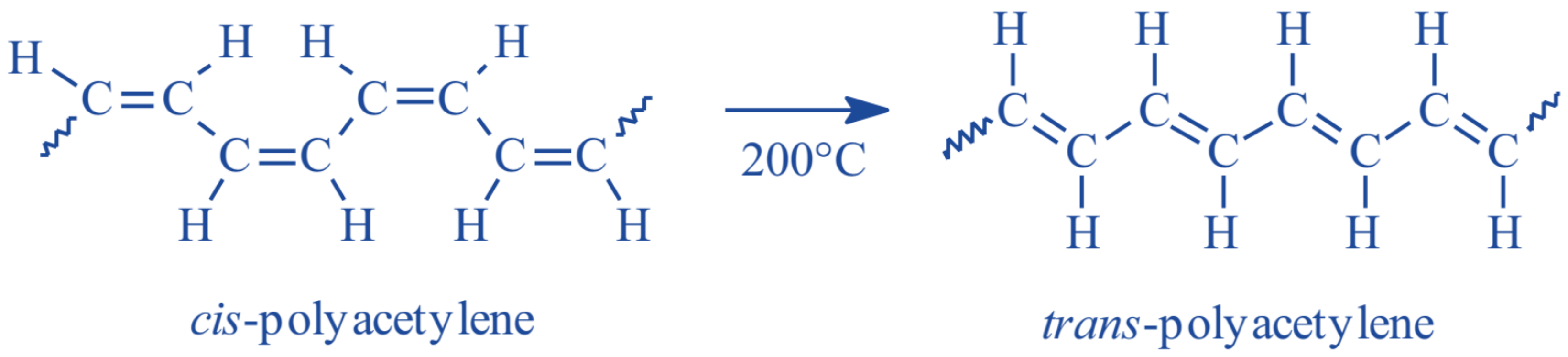

Example of Conducting Polymer



trans-polyacetylene

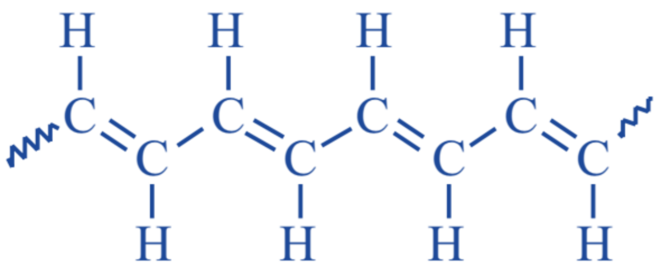
Conductive Polymer

- Polyacetylene with molecular weight up to 1 million can be prepared through a complicated process involving a **metathesis polymerization**.
- The *cis* isomer of polyacetylene can be transformed to the more stable *trans* isomer by heating, as follows:



- The *trans* isomer has higher conductivity (4.4×10^{-5} S cm^{-1}) than the *cis* isomer (1.7×10^{-9} S cm^{-1}).

What causes polyacetylene to be electrically conductive?



trans-polyacetylene

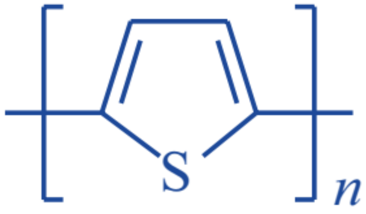
"Polyacetylene and other conductive organic polymers conduct electricity *due to the extended conjugation of π-electron clouds*.

Alternately stating, they have delocalized π-electron clouds in long range".

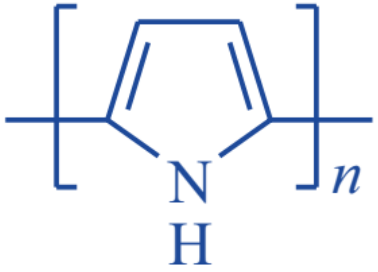
As a result the electrons are mobile along the main chain of the polymer.

Other Conductive Polymer

Poly(*p*-phenylene) (PPP)



Polythiophene (PT)



Polypyrrole (PPy)

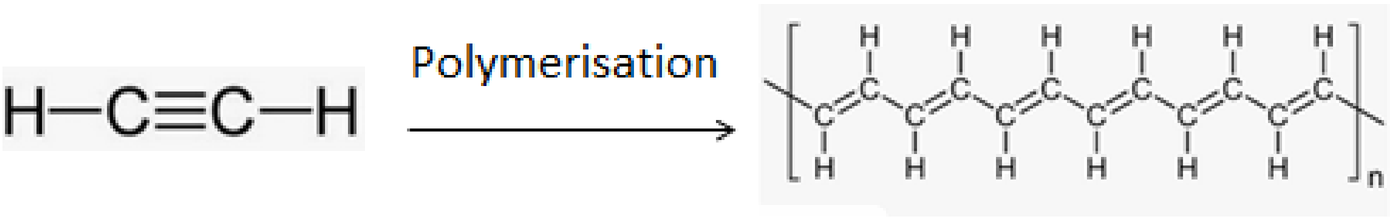
Conducting polymer structures

Polymers with alternating double bonds can conduct electricity, hence called conducting polymers

Examples for conducting polymers

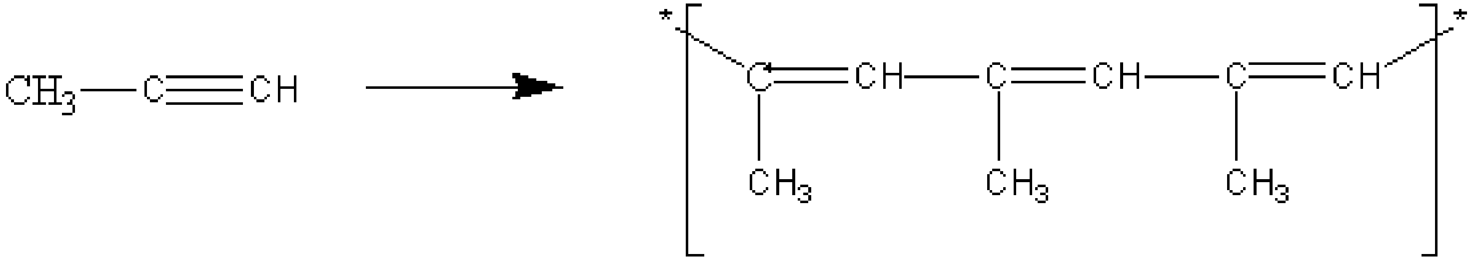
monomers

polymer



acetylene

polyacetylene



Conducting polymer structures

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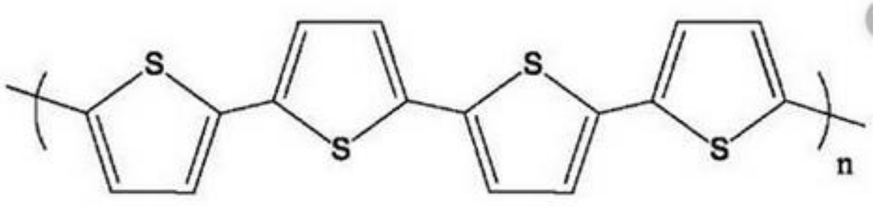
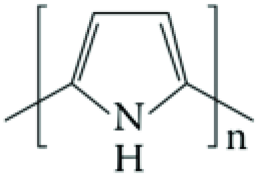
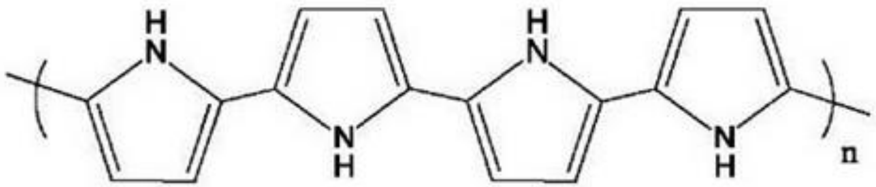
Examples for conducting polymers



polythiophene





polypyrrole

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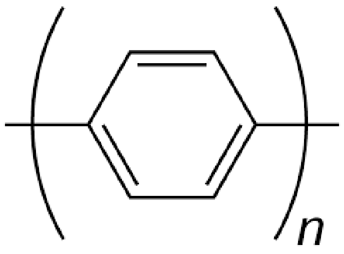
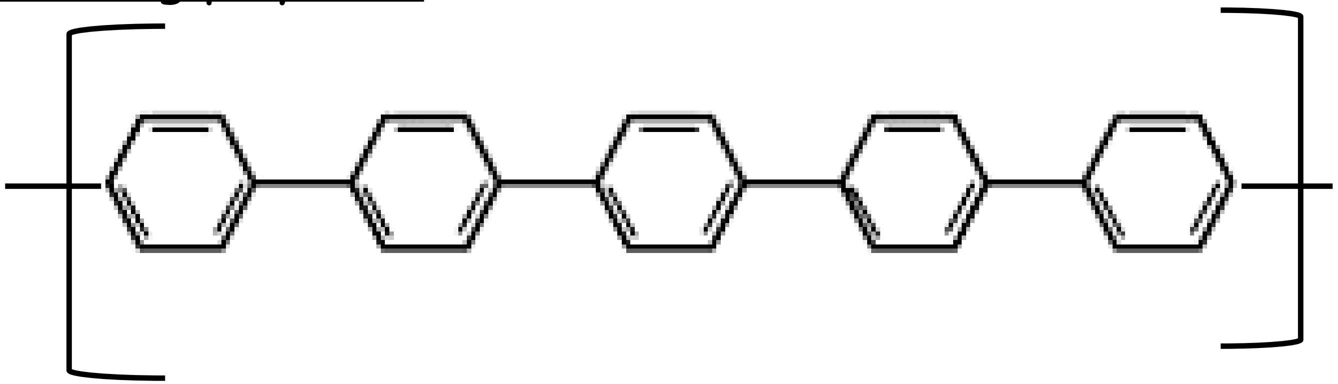
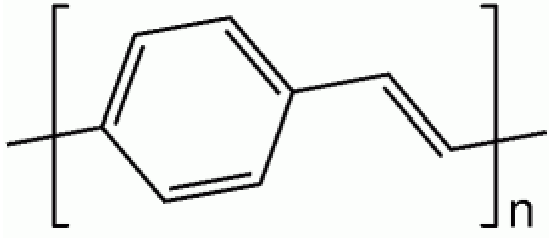
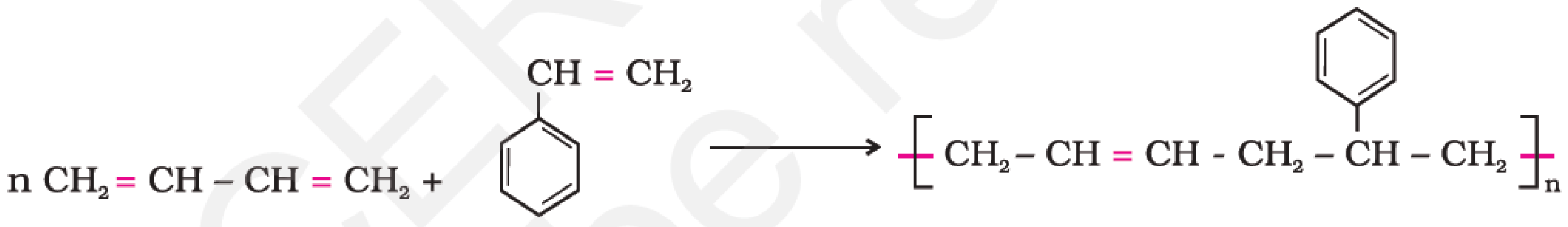






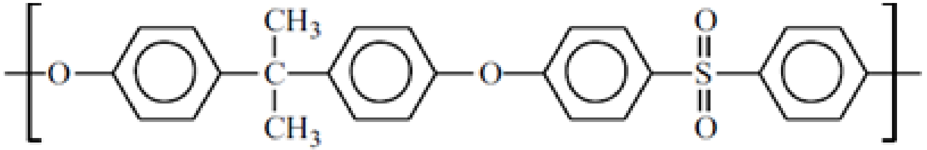
Poly(*p*-phenylene vinylene)

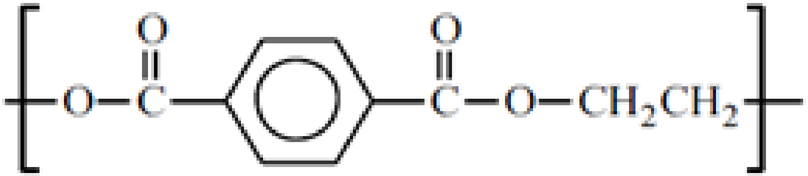
Non-Conducting polymer structures



1, 3-Butadiene | Styrene | Butadiene - styrene copolymer

non-conducting polymer
because no conjugation
or alternate double bonds





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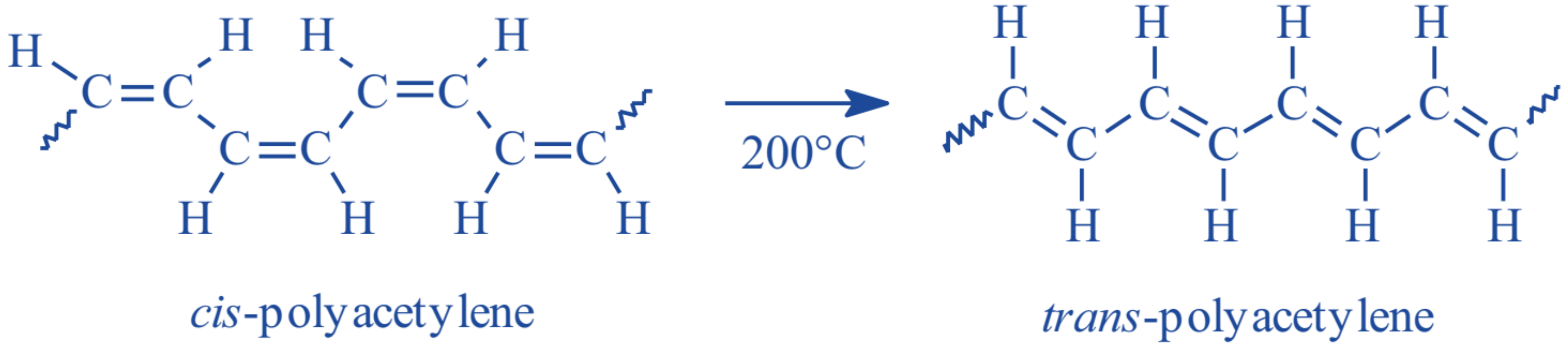

- The ***trans* isomer** has higher conductivity (4.4×10^{-5} S cm^{-1}) than the ***cis* isomer** (1.7×10^{-9} S cm^{-1}).

QN **Which one of the following is a conductive polymer?**